

Enterprise Network Design and Administration  
Building a Secure Enterprise LAN using  
Windows Server 2022 and Cisco Packet Tracer

**Project Documentation**

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Network Engineering Portfolio

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# **1.Executive Summary**

The Enterprise Network Design and Administration project was developed to simulate a real-world enterprise network environment using Windows Server 2022 and Cisco Packet Tracer. The primary objective of this project was to design, configure, and validate a secure, scalable, and centrally managed network infrastructure that reflects common practices used in modern organizations.

The project demonstrates the deployment of Windows Server 2022 as the primary Domain Controller with Active Directory Domain Services (AD DS), Domain Name System (DNS), Dynamic Host Configuration Protocol (DHCP), and Group Policy Objects (GPO). Organizational Units (OUs), domain users, security groups, shared folders, and NTFS permissions were configured to implement centralized identity management, resource access control, and administrative policies.

On the networking side, Cisco Packet Tracer was used to design the enterprise network topology. Virtual Local Area Networks (VLANs) were implemented to logically separate server and client networks, while Router-on-a-Stick technology enabled inter-VLAN communication. Additional network services such as trunking, IP addressing, DHCP integration, and network connectivity testing were configured to ensure reliable communication between network devices.

Comprehensive validation was performed using connectivity tests, DNS resolution, DHCP lease verification, Active Directory authentication, Group Policy enforcement, and shared resource accessibility. The successful completion of these tests confirmed that the implemented infrastructure operates as an integrated enterprise network.

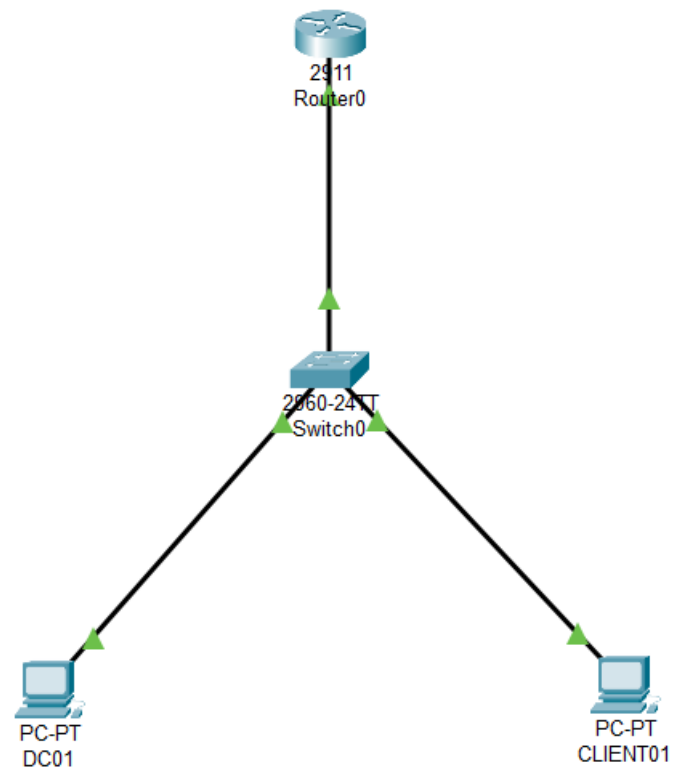
This project strengthened practical skills in Windows Server administration, enterprise network design, Cisco networking fundamentals, troubleshooting, and system documentation. It serves as a portfolio project demonstrating the ability to design, implement, secure, and validate an enterprise network environment using industry-relevant technologies.

## **2. Project Objective**

- To design and implement a secure and scalable enterprise network infrastructure using Windows Server 2022 and Cisco Packet Tracer.
- To deploy and configure Active Directory Domain Services (AD DS) for centralized authentication, authorization, and user management.
- To configure Domain Name System (DNS) services for reliable hostname resolution within the enterprise network.
- To implement Dynamic Host Configuration Protocol (DHCP) for automatic IP address allocation and centralized network configuration.
- To create and manage Organizational Units (OUs), domain users, security groups, and administrative policies using Active Directory.
- To configure shared folders and implement NTFS and Share permissions for secure resource access.
- To design a segmented network using Virtual Local Area Networks (VLANs) to improve security and network organization.
- To configure trunk links and Router-on-a-Stick for communication between multiple VLANs.
- To implement Group Policy Objects (GPOs) for centralized management of user and computer configurations.
- To perform connectivity, DNS, DHCP, Active Directory, and resource access testing to validate the functionality of the enterprise network.
- To develop troubleshooting skills by identifying and resolving configuration and connectivity issues encountered during implementation.
- To document the complete network architecture, configuration process, testing methodology, and project outcomes following professional engineering documentation practices.

### **3. Technology Used**

| <b>Technology / Software</b>               | <b>Version</b>         | <b>Purpose</b>                                                                                                                                                              |
|--------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Windows Server 2022                        | 2022                   | Configured as the Domain Controller to provide Active Directory Domain Services (AD DS), DNS, DHCP, Group Policy, and centralized user management.                          |
| Windows 10/11 Client                       | Windows 10/11          | Joined to the Active Directory domain to verify authentication, Group Policy application, DNS resolution, DHCP configuration, and resource access.                          |
| VMware Workstation Pro                     | Latest Stable Version  | Used to create and manage virtual machines for the Windows Server and client systems.                                                                                       |
| Cisco Packet Tracer                        | 8.x                    | Used to design the enterprise network topology, configure routers and switches, implement VLANs, trunking, and inter-VLAN routing.                                          |
| Active Directory Domain Services (AD DS)   | Windows Server Role    | Used for centralized authentication, authorization, user management, and domain administration.                                                                             |
| Domain Name System (DNS)                   | Windows Server Role    | Provided hostname resolution for devices and services within the enterprise network.                                                                                        |
| Dynamic Host Configuration Protocol (DHCP) | Windows Server Role    | Automatically assigned IP addresses and network configuration to client devices.                                                                                            |
| Group Policy Management (GPO)              | Windows Server Feature | Applied centralized security settings, user restrictions, and administrative policies across the domain.                                                                    |
| NTFS & Share Permissions                   | Windows Server         | Controlled access to shared folders and implemented role-based resource permissions.                                                                                        |
| Cisco Router                               | Cisco Packet Tracer    | Configured Router-on-a-Stick, inter-VLAN routing, DHCP relay (if implemented), and gateway services.                                                                        |
| Cisco Switch                               | Cisco Packet Tracer    | Configured VLANs, access ports, trunk ports, and Layer-2 switching between network devices.                                                                                 |
| Command Prompt (CMD)                       | Windows                | Used for testing network connectivity, DNS resolution, DHCP configuration, and domain communication using commands such as ping, ipconfig, nslookup, gpresult, and net use. |



***Figure 4.1: Enterprise Network Topology Designed in Cisco Packet Tracer***

## 4. System Requirements

### Hardware Requirements

| Component | Minimum Requirement                                    | Recommended                 |
|-----------|--------------------------------------------------------|-----------------------------|
| Processor | Intel Core i5 (6th Generation or Higher) / AMD Ryzen 5 | Intel Core i7 / AMD Ryzen 7 |
| RAM       | 8 GB                                                   | 16 GB or Higher             |
| Storage   | 50 GB Free Disk Space                                  | 100 GB SSD                  |
| Network   | Standard Ethernet/Wi-Fi Adapter                        | Gigabit Ethernet            |

### Software Requirements

| Software                                   | Purpose                                                                 |
|--------------------------------------------|-------------------------------------------------------------------------|
| Windows 11                                 | Host Operating System                                                   |
| VMware Workstation Pro                     | Virtualization platform for Windows Server and Client virtual machines  |
| Windows Server 2022                        | Domain Controller providing AD DS, DNS, DHCP, and Group Policy services |
| Windows 10/11                              | Client operating system joined to the Active Directory domain           |
| Cisco Packet Tracer (Version 8.x or later) | Network topology design and Cisco router/switch configuration           |
| Microsoft Remote Desktop (Optional)        | Remote administration of virtual machines                               |
| Command Prompt (CMD)                       | Network troubleshooting and verification                                |
| PowerShell                                 | Windows Server administration and management                            |
| Active Directory Users and Computers       | User, Group, and Organizational Unit management                         |
| DNS Manager                                | DNS zone and record management                                          |
| DHCP Manager                               | DHCP scope and lease management                                         |
| Group Policy Management Console (GPMC)     | Creation and administration of Group Policy Objects                     |

### Network Requirements

- Domain Name: **lab.local**
- Domain Controller IP Address: **192.168.10.10**
- DNS Server: **192.168.10.10**
- DHCP Scope: **192.168.10.100 – 192.168.10.200**
- Default Gateway: Configured on the Cisco Router
- VLAN 10: Server Network

- VLAN 20: Client Network

## Virtual Machines Used

| Virtual Machine | Operating System    | Purpose                                                                                  |
|-----------------|---------------------|------------------------------------------------------------------------------------------|
| DC01            | Windows Server 2022 | Active Directory Domain Controller, DNS, DHCP, Group Policy                              |
| CLIENT01        | Windows 10/11       | Domain-Joined Client used for testing authentication, policies, and network connectivity |

## Network Devices

| Device            | Quantity | Purpose                                |
|-------------------|----------|----------------------------------------|
| Cisco Router      | 1        | Inter-VLAN Routing and Default Gateway |
| Cisco 2960 Switch | 1        | VLAN and Layer-2 Switching             |
| Windows Server    | 1        | Domain Services                        |
| Windows Client    | 1        | Enterprise Client                      |

## 6. Network Architecture

### Overview

The enterprise network architecture was designed to simulate a real-world organizational environment using Windows Server 2022 and Cisco Packet Tracer. The infrastructure provides centralized user authentication, network management, IP address allocation, hostname resolution, and secure communication between multiple network segments. The network consists of a Windows Server 2022 Domain Controller (DC01), a domain-joined Windows client (CLIENT01), a Cisco router, and a Cisco Layer-2 switch. The Windows Server hosts Active Directory Domain Services (AD DS), DNS, DHCP, and Group Policy services, while the Cisco devices provide VLAN segmentation and inter-VLAN routing.

The network is logically divided into two Virtual Local Area Networks (VLANs):

- **VLAN 10 (Servers):** Hosts the Windows Server 2022 Domain Controller and enterprise services such as Active Directory, DNS, and DHCP.
- **VLAN 20 (Clients):** Hosts client computers that obtain network configuration automatically, join the Active Directory domain, and access centralized enterprise resources.

Communication between VLANs is achieved using the Router-on-a-Stick technique, where a single physical connection between the router and switch carries multiple VLANs through IEEE 802.1Q trunking. This allows clients in different VLANs to communicate securely while maintaining logical network separation.

The overall architecture follows common enterprise networking practices by combining centralized directory services with VLAN-based network segmentation, resulting in improved scalability, security, and ease of administration.



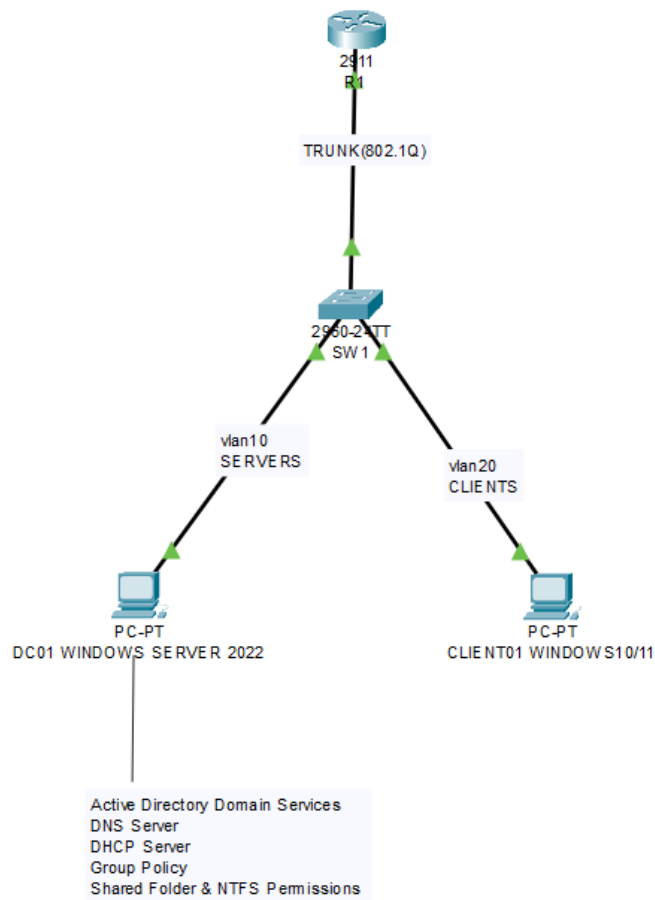


Figure 6.1: Enterprise Network Architecture

## Network Components

The enterprise network consists of the following components:

### Cisco Router (R1):

Acts as the default gateway for all VLANs and performs inter-VLAN routing using Router-on-a-Stick. The router also provides network connectivity between different logical segments.

### Cisco Layer-2 Switch (SW1):

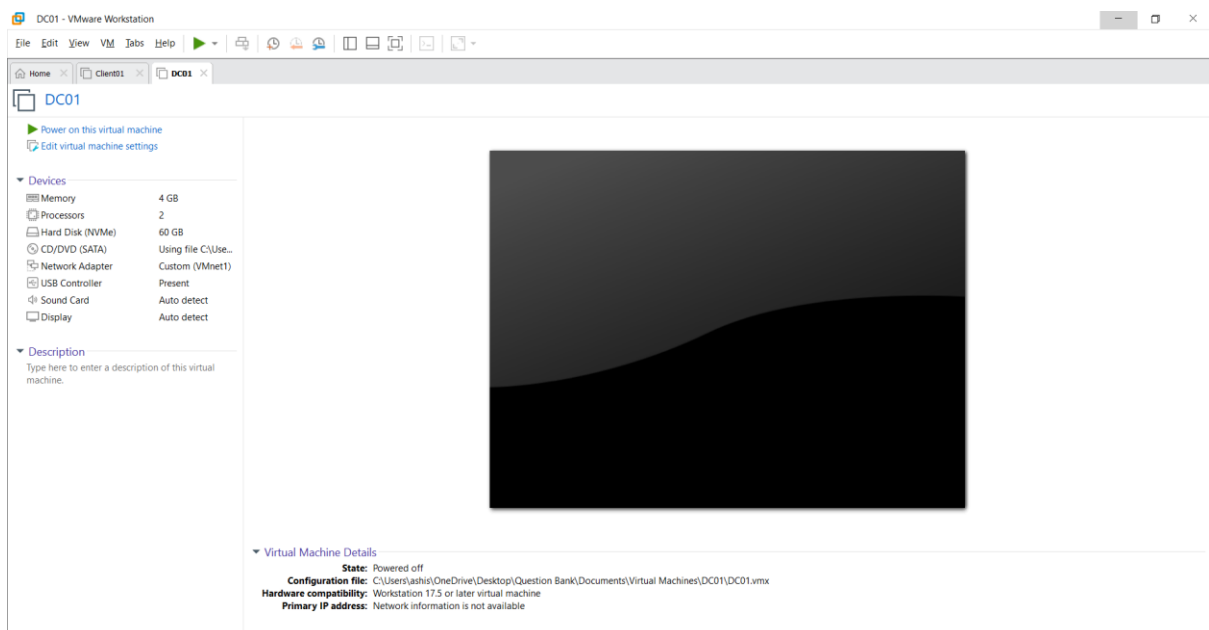
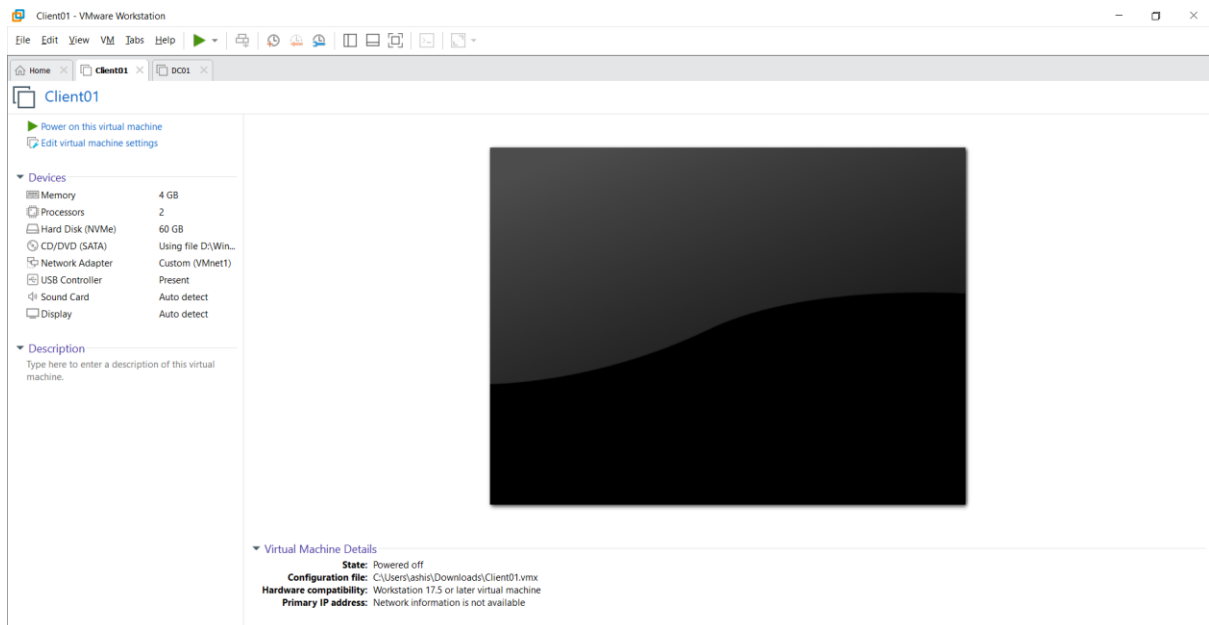
Connects all end devices and implements VLAN segmentation. The switch is configured with access ports for end devices and a trunk port connecting to the router.

### Windows Server 2022 (DC01):

Functions as the Domain Controller and hosts Active Directory Domain Services (AD DS), DNS, DHCP, Group Policy Management, Organizational Units, user accounts, security groups, and shared folders.

### Windows Client (CLIENT01):

Represents an enterprise workstation that joins the Active Directory domain, receives network configuration through DHCP, authenticates using domain credentials, and accesses shared enterprise resources.



**Figure 6.2,6.3 VMware Workstation**

# IP Addressing Scheme

## Network Addressing Plan

The enterprise network is divided into two VLANs, each assigned a dedicated IPv4 subnet. VLAN 10 is used for server infrastructure, while VLAN 20 is designated for client systems. A Cisco router performs inter-VLAN routing using the Router-on-a-Stick technique, allowing communication between the two networks.

## IP Address Allocation

| Device                     | Interface | VLAN    | IP Address            | Subnet Mask   | Default Gateway |
|----------------------------|-----------|---------|-----------------------|---------------|-----------------|
| Router (R1)                | G0/0.10   | VLAN 10 | 192.168.10.1          | 255.255.255.0 | N/A             |
| Router (R1)                | G0/0.20   | VLAN 20 | 192.168.20.1          | 255.255.255.0 | N/A             |
| DC01 (Windows Server 2022) | Ethernet  | VLAN 10 | 192.168.10.10         | 255.255.255.0 | 192.168.10.1    |
| CLIENT01                   | Ethernet  | VLAN 20 | DHCP / 192.168.20.100 | 255.255.255.0 | 192.168.20.1    |
| DNS Server                 | DC01      | VLAN 10 | 192.168.10.10         | 255.255.255.0 | -               |
| DHCP Server                | DC01      | VLAN 10 | 192.168.10.10         | 255.255.255.0 | -               |

## DHCP Scope

| Parameter       | Value                           |
|-----------------|---------------------------------|
| Scope Name      | LabScope                        |
| Network Address | 192.168.20.0                    |
| Subnet Mask     | 255.255.255.0                   |
| Address Pool    | 192.168.20.100 – 192.168.20.200 |
| Default Gateway | 192.168.20.1                    |
| DNS Server      | 192.168.10.10                   |

| Parameter   | Value     |
|-------------|-----------|
| Domain Name | lab.local |

### VLAN Information

| VLAN ID | VLAN Name | Purpose                          | Network         |
|---------|-----------|----------------------------------|-----------------|
| 10      | SERVERS   | Windows Server, AD DS, DNS, DHCP | 192.168.10.0/24 |
| 20      | CLIENTS   | Domain-Joined Client Computers   | 192.168.20.0/24 |

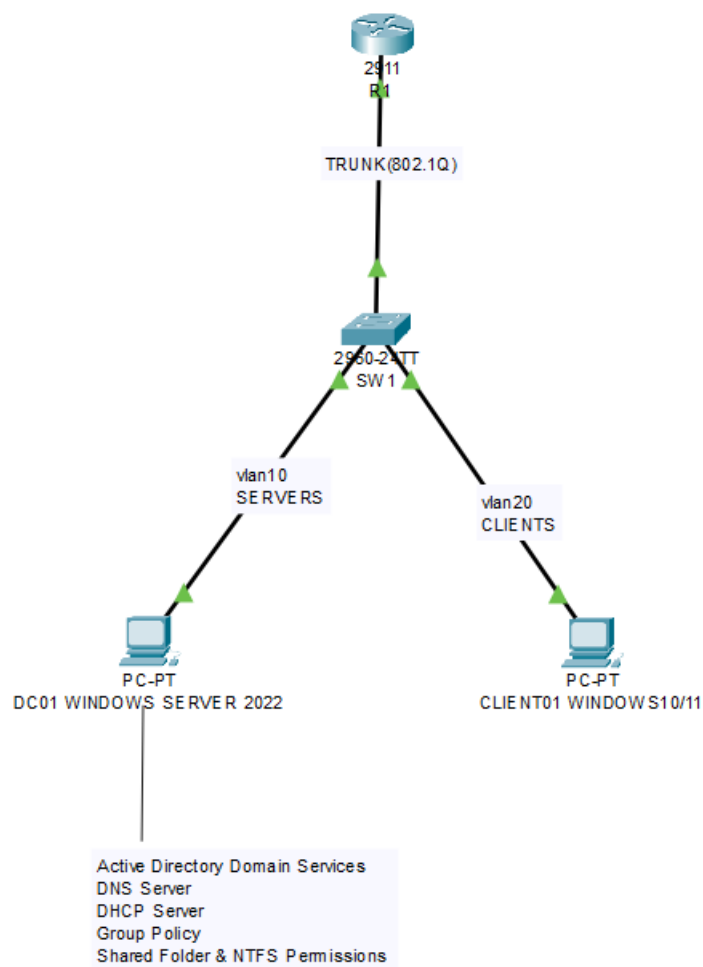
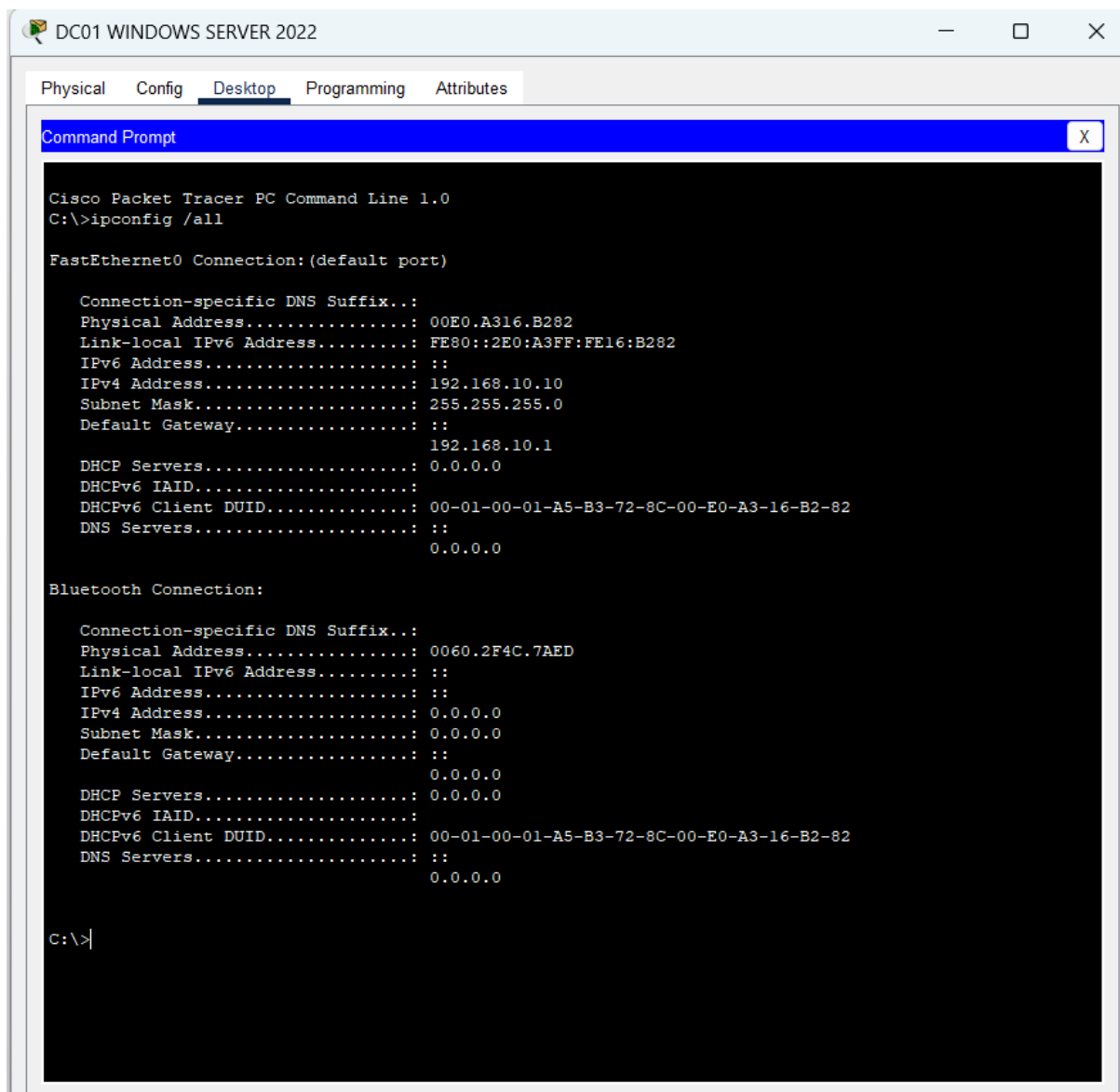
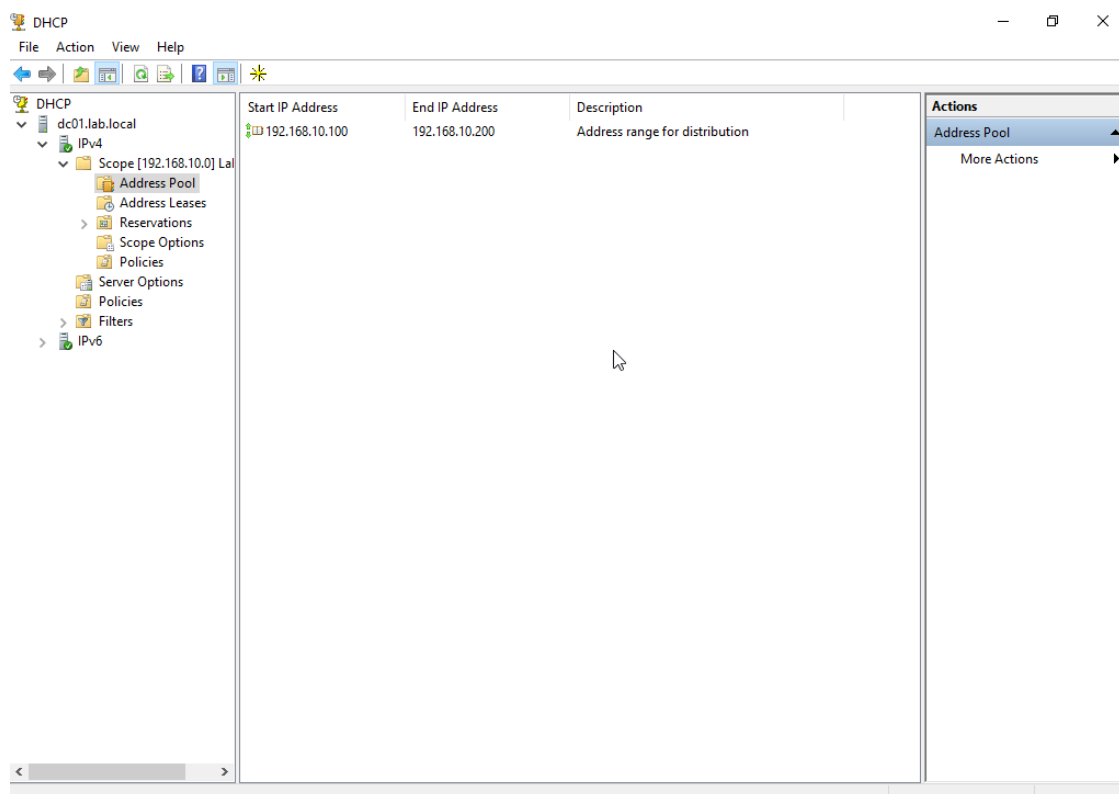
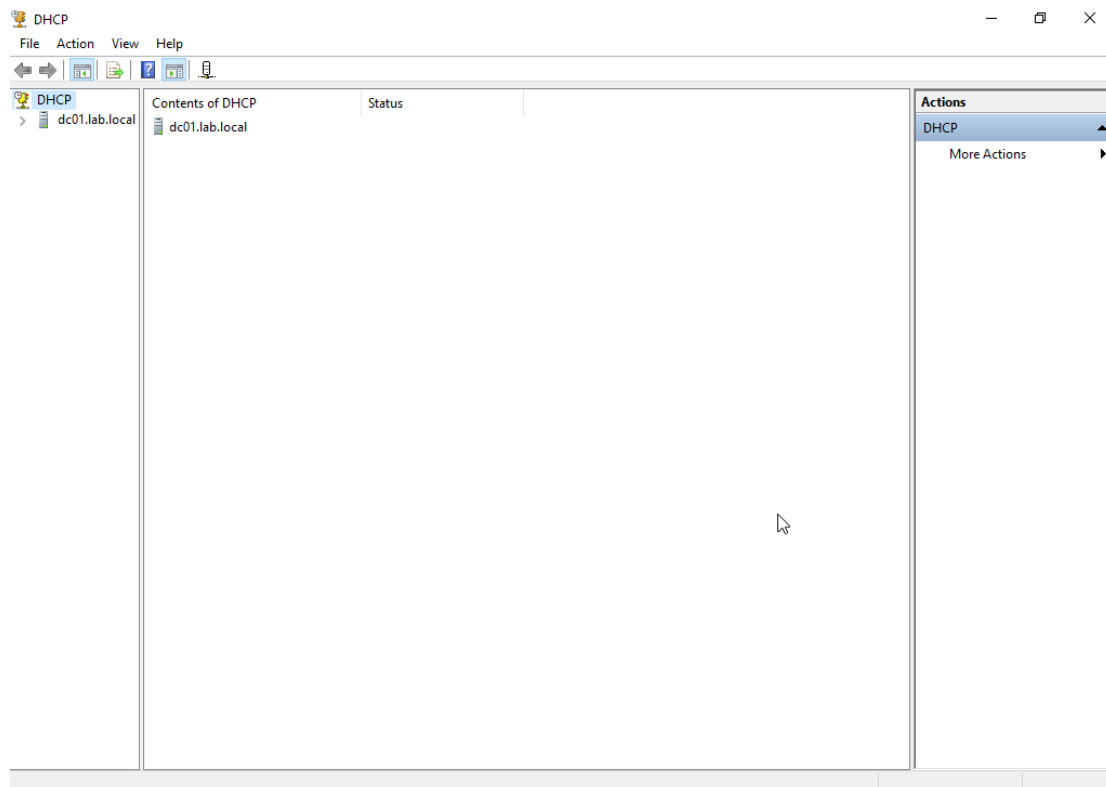
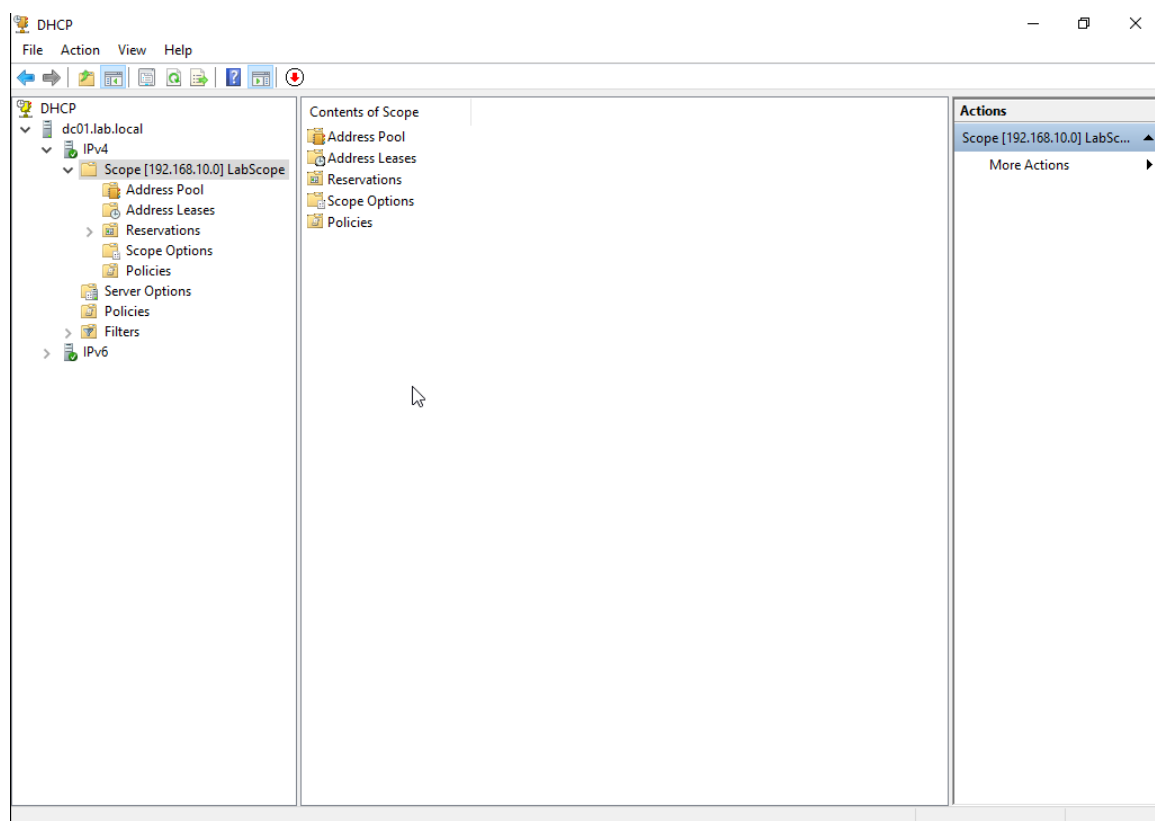
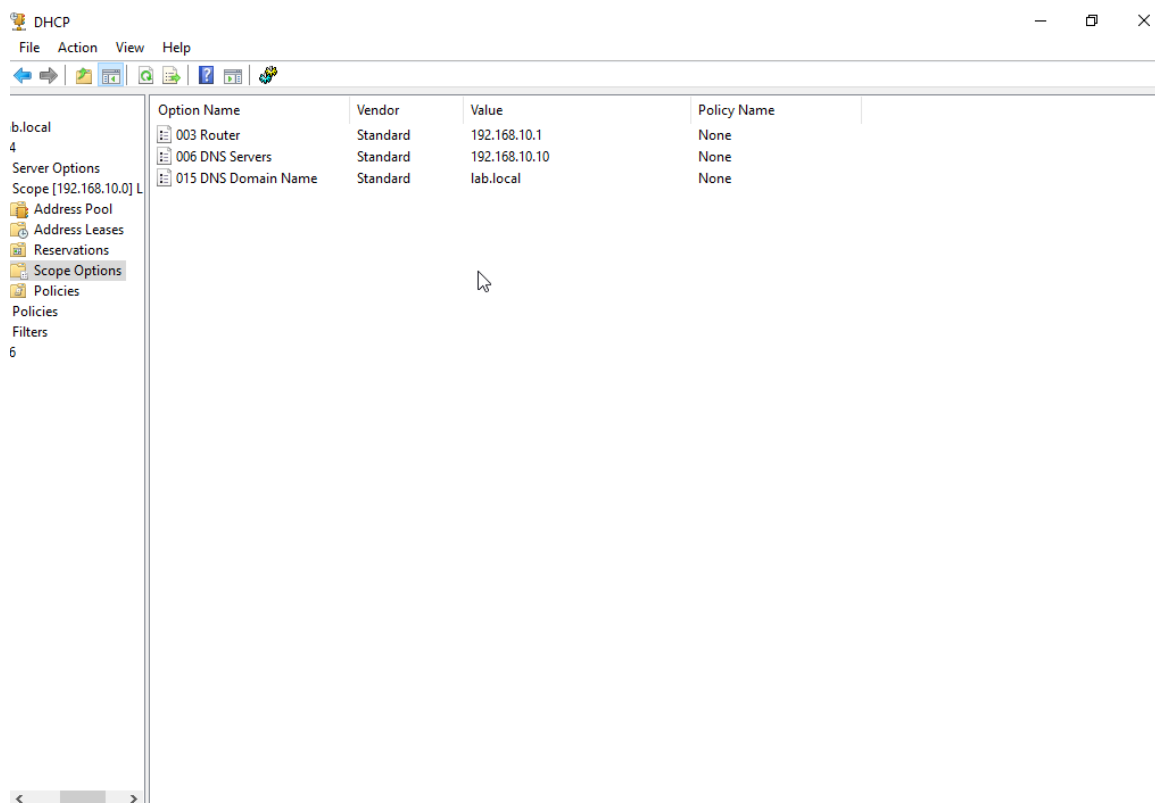


Figure 7.1 Network Topology

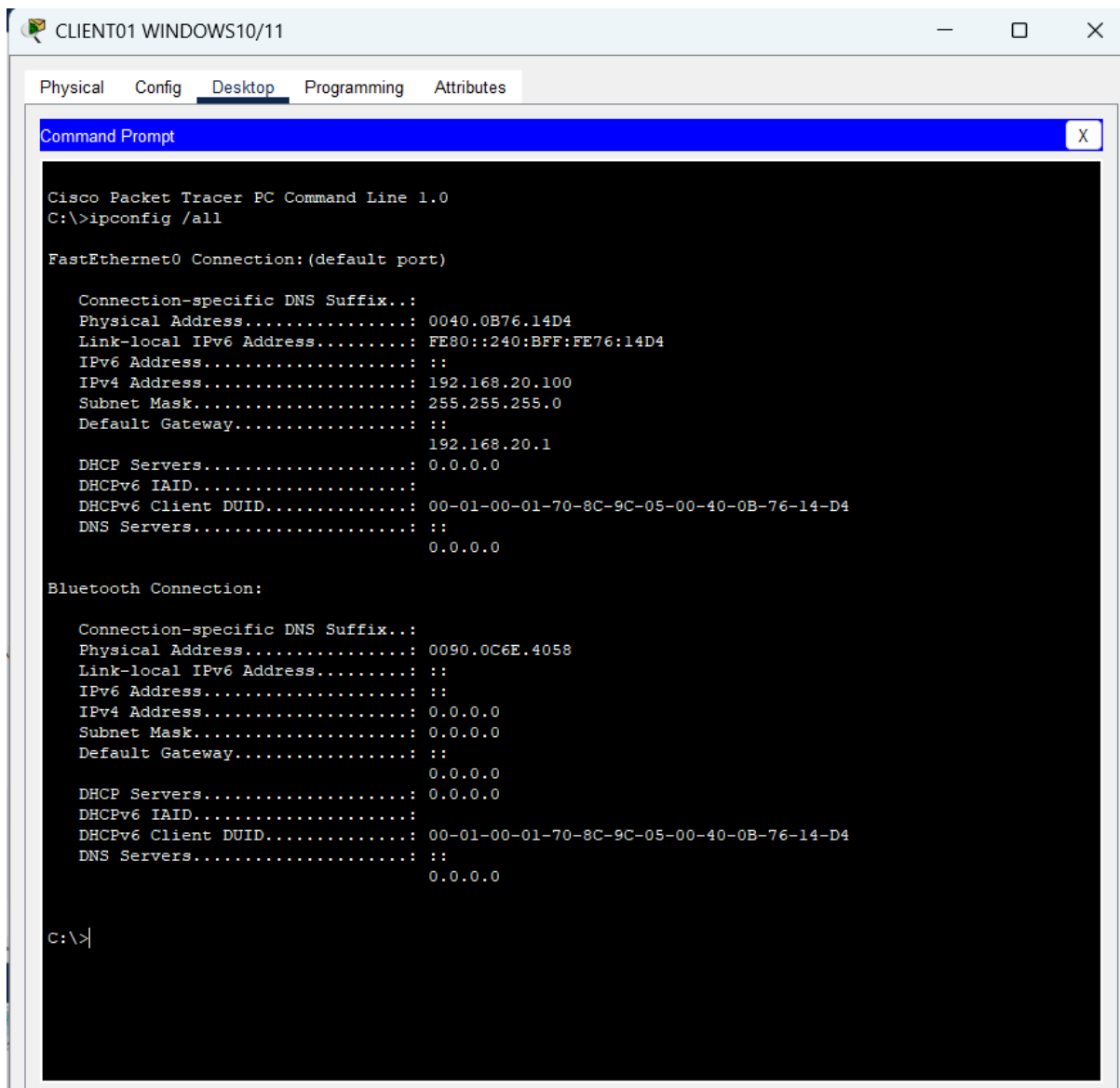


*Figure 7.2: DC01 showing the server IP configuration*





**Figure 7.3: DHCP Manager showing the active scope (LabScope) and address pool.**



**Figure 7.4:** CLIENT01 showing the client IP configuration



# Windows Server Configuration

## Server Deployment

A virtual machine named **DC01** was created using VMware Workstation Pro and installed with Windows Server 2022. The server was configured with a static IPv4 address to ensure consistent network communication and reliable operation of enterprise services.

## Initial Network Configuration

The following network settings were assigned to the server:

| Parameter            | Value               |
|----------------------|---------------------|
| Computer Name        | DC01                |
| Operating System     | Windows Server 2022 |
| IPv4 Address         | 192.168.10.10       |
| Subnet Mask          | 255.255.255.0       |
| Default Gateway      | 192.168.10.1        |
| Preferred DNS Server | 192.168.10.10       |
| Domain Name          | lab.local           |

After verifying the network configuration, the server was promoted to a Domain Controller by installing the Active Directory Domain Services (AD DS) role and creating a new Active Directory forest named **lab.local**.

## Installed Server Roles

The following Windows Server roles were installed and configured:

- Active Directory Domain Services (AD DS)
- Domain Name System (DNS)
- Dynamic Host Configuration Protocol (DHCP)
- Group Policy Management

These services provide centralized authentication, name resolution, automatic IP address assignment, and policy-based management throughout the enterprise network.

## Active Directory Deployment

A new Active Directory forest named **lab.local** was created during the server promotion process. Organizational Units (OUs), domain users, and security groups were configured to organize administrative resources and simplify access management.

## DNS Configuration

The DNS Server role was configured to host the **lab.local** forward lookup zone. Host (A), Alias (CNAME), and Pointer (PTR) records were created to enable hostname resolution within the enterprise environment.

## DHCP Configuration

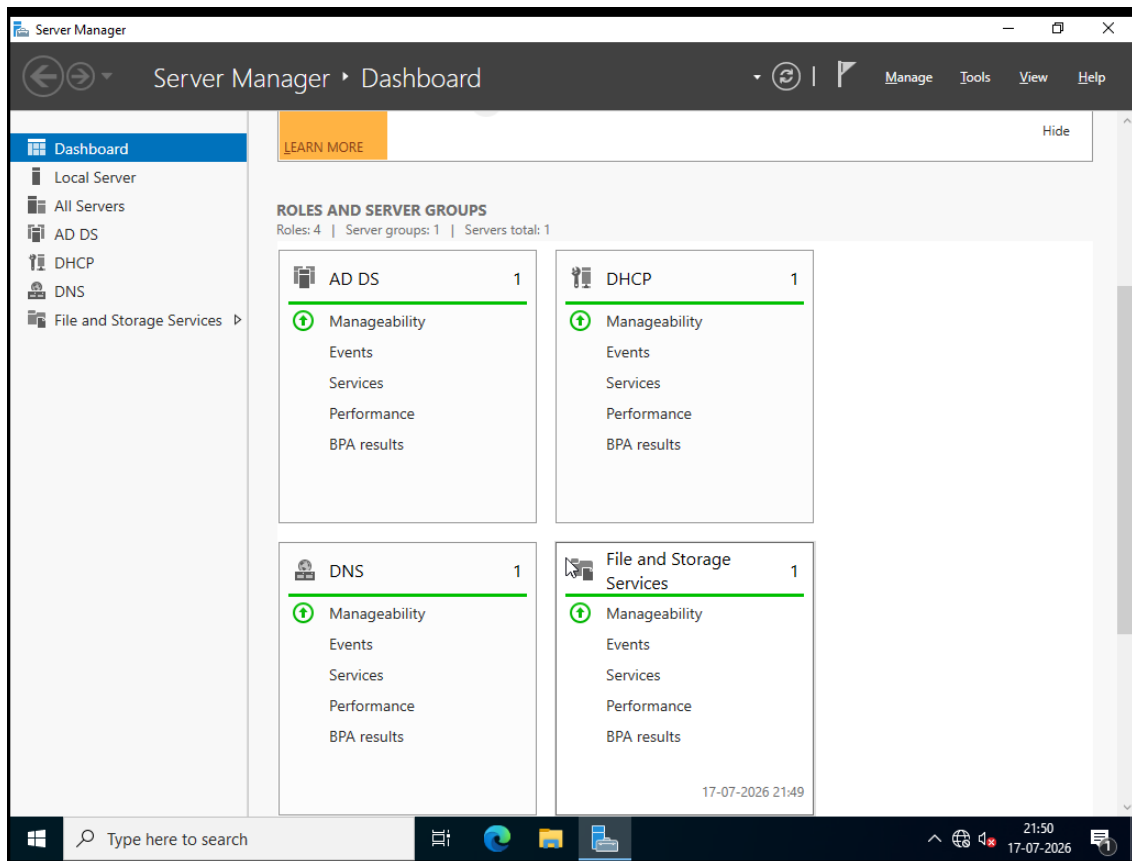
The DHCP Server role was configured to automatically assign IPv4 addresses to client systems. A DHCP scope named **LabScope** was created, defining the address pool, subnet mask, default gateway, DNS server, and domain name distributed to clients.

## Group Policy Configuration

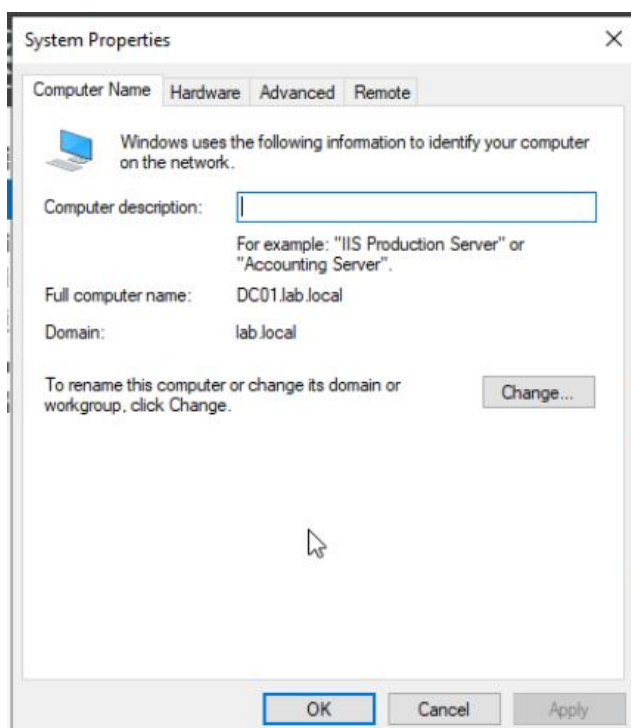
Group Policy Objects (GPOs) were created to centrally manage user and computer settings across the domain. Administrative restrictions and security policies were successfully applied to domain-joined client systems.

## Shared Resource Configuration

A shared folder was created on the server with both Share Permissions and NTFS Permissions configured. Access to shared resources was controlled through Active Directory security groups, demonstrating role-based access management within the enterprise network.



**Figure 8.1 Server Manager Dashboard**



**Figure 8.2 System Properties**

```
Select Administrator: Command Prompt

C:\Users\Administrator>ipconfig /all

Windows IP Configuration

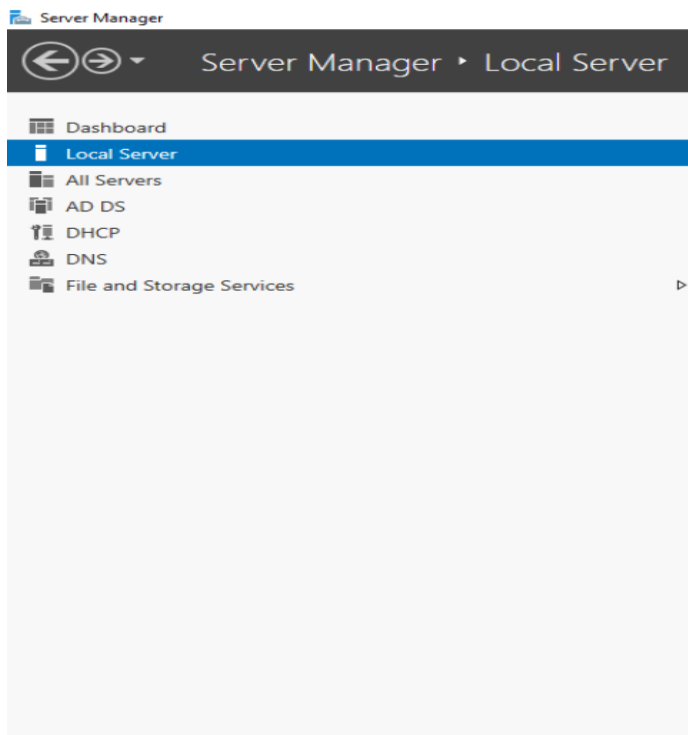
    Host Name . . . . . : DC01
    Primary Dns Suffix . . . . . : lab.local
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No
    DNS Suffix Search List. . . . . : lab.local

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix . : 
    Description . . . . . : Intel(R) 82574L Gigabit Network Connection
    Physical Address. . . . . : 00-0C-29-7F-12-46
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::cd82:f7dc:a85e:208%6(Preferred)
    IPv4 Address. . . . . : 192.168.10.10(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 
    DHCPv6 IAID . . . . . : 100666409
    DHCPv6 Client DUID. . . . . : 00-01-00-01-31-A7-A1-A2-00-0C-29-7F-12-46
    DNS Servers . . . . . : ::1
                           192.168.10.10
    NetBIOS over Tcpip. . . . . : Enabled

C:\Users\Administrator>
```

*Figure 8.3 Shows Static IP, DNS Server, Gateway*



*Figure 8.4 Add Roles and Features*

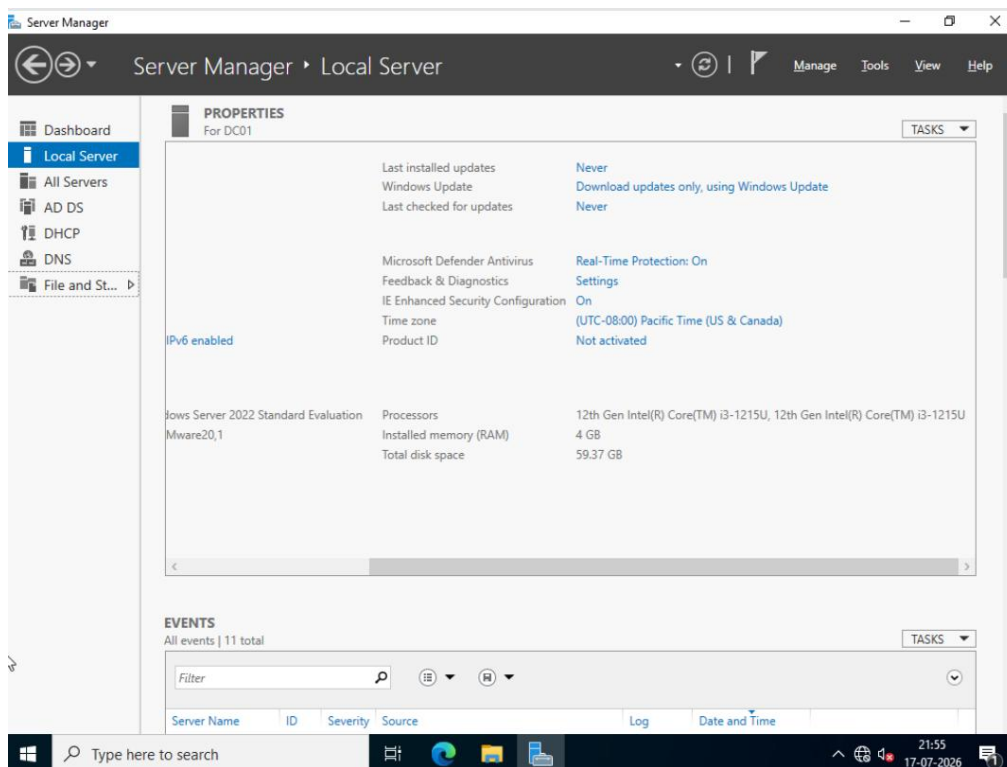
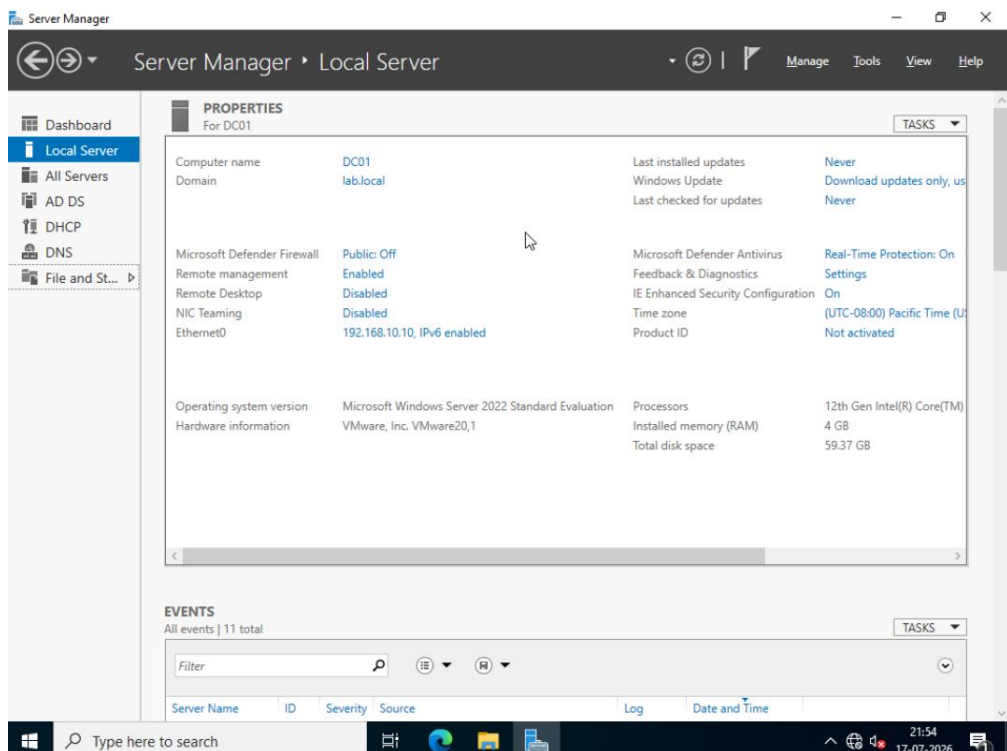


Figure 8.5 Local Server

# Active Directory Configuration

Active Directory Domain Services (AD DS) was implemented to provide centralized identity management, authentication, authorization, and administration within the enterprise network. By deploying a domain controller, users, computers, and network resources can be managed from a single location, improving security, scalability, and administrative efficiency.

## Overview

Active Directory Domain Services (AD DS) was deployed on the Windows Server 2022 system (DC01) to establish a centralized directory service for the enterprise network. The server was promoted to a Domain Controller by creating a new Active Directory forest named **lab.local**. The Active Directory environment enables centralized management of users, computers, security groups, and organizational resources while providing secure authentication and authorization throughout the domain.

---

## Domain Information

| Parameter               | Value                         |
|-------------------------|-------------------------------|
| Domain Name             | lab.local                     |
| Domain Controller       | DC01                          |
| Forest Functional Level | Windows Server 2022 (Default) |
| Domain Functional Level | Windows Server 2022 (Default) |

---

## Organizational Units (OUs)

To organize administrative resources efficiently, Organizational Units (OUs) were created within the domain. OUs allow administrators to logically group users, computers, and security groups while enabling the application of Group Policies to specific departments.

Example Organizational Units:

- IT
- HR
- Finance
- Computers
- Servers

---

## Domain Users

Domain user accounts were created to simulate employees within the organization. Each user account is centrally managed through Active Directory, allowing authentication from any domain-joined computer.

Example User Accounts:

Username Department

Ashish IT

TestUser HR

---

## Security Groups

Security groups were created to simplify permission management for shared resources and administrative tasks.

#### Example Groups:

| Group Name    | Purpose                       |
|---------------|-------------------------------|
| IT_Users      | Access to IT shared resources |
| HR_Users      | Access to HR resources        |
| Domain Admins | Administrative privileges     |

Users were added to the appropriate security groups based on their roles within the organization.

---

### Domain Join

The client computer (CLIENT01) was successfully joined to the **lab.local** domain. After joining the domain, users were able to authenticate using their domain credentials and receive centralized policies from the Domain Controller.

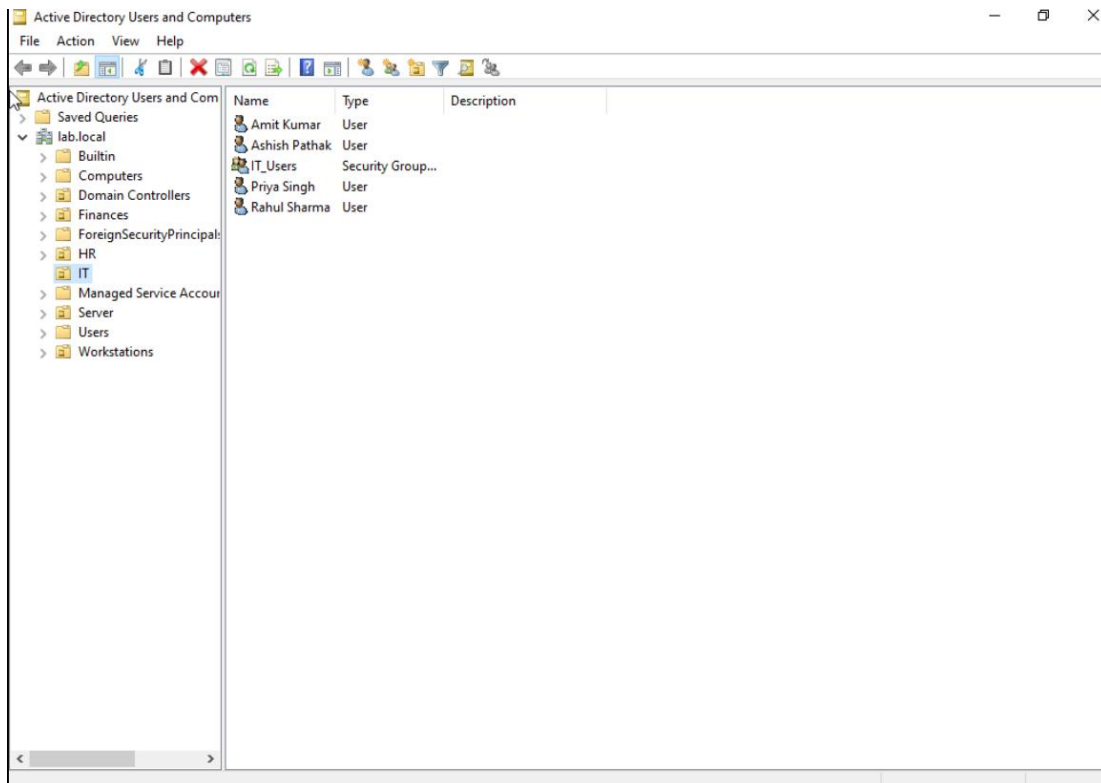
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### Active Directory Features Implemented

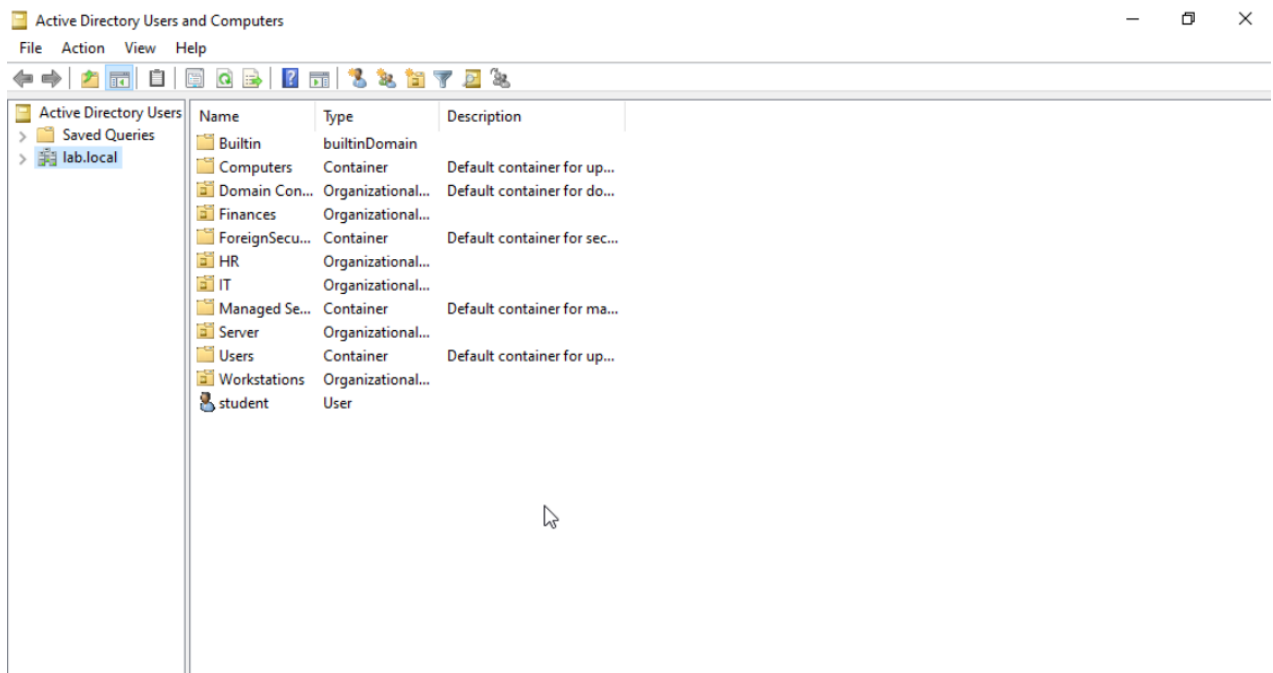
The following Active Directory features were successfully configured:

- Active Directory Domain Services (AD DS)
- Organizational Units (OUs)
- Domain User Accounts
- Security Groups
- Domain Authentication
- Centralized User Management
- Group Membership Management
- Domain-Joined Client Management

The successful implementation of these features provides centralized administration and forms the foundation for DNS, DHCP, Group Policy, and secure resource access throughout the enterprise network.

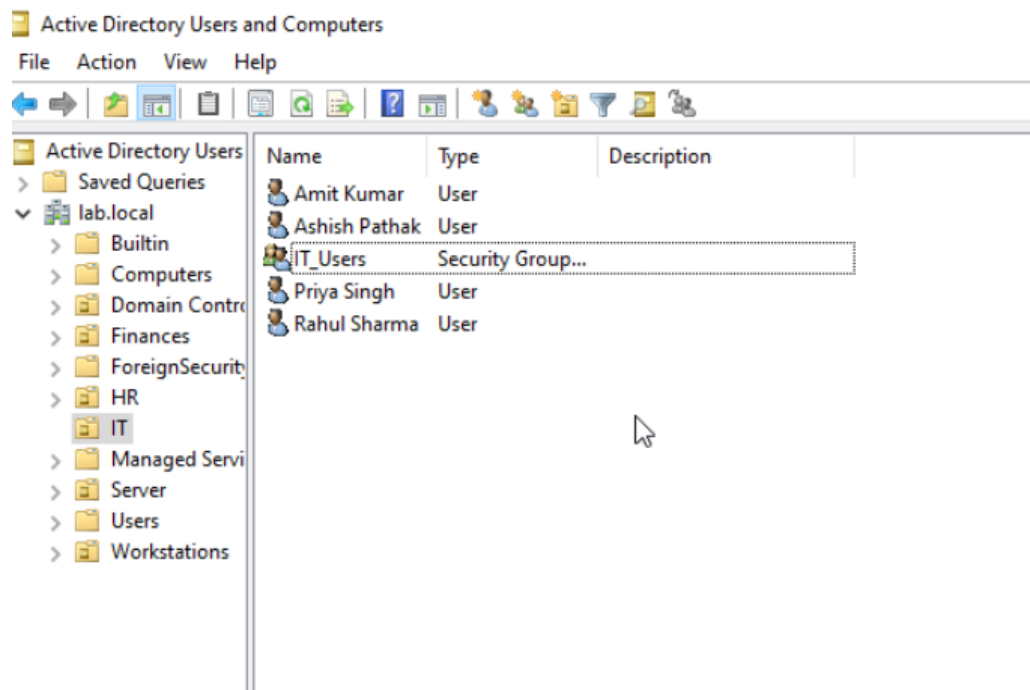


**Figure 9.1: Active Directory Users and Computers Console.**

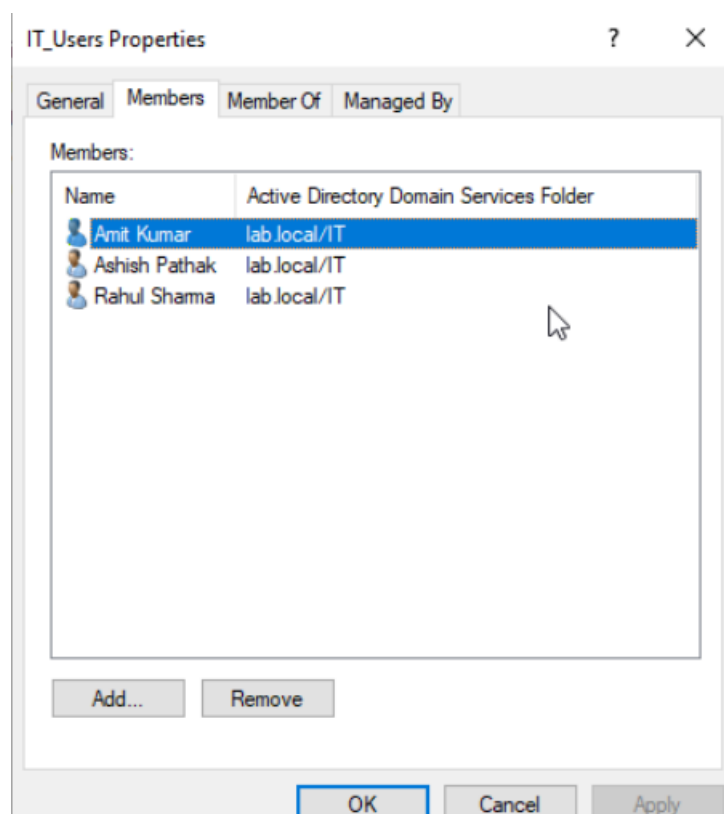


**Figure 9.2: Organizational Units created within the lab.local domain.**

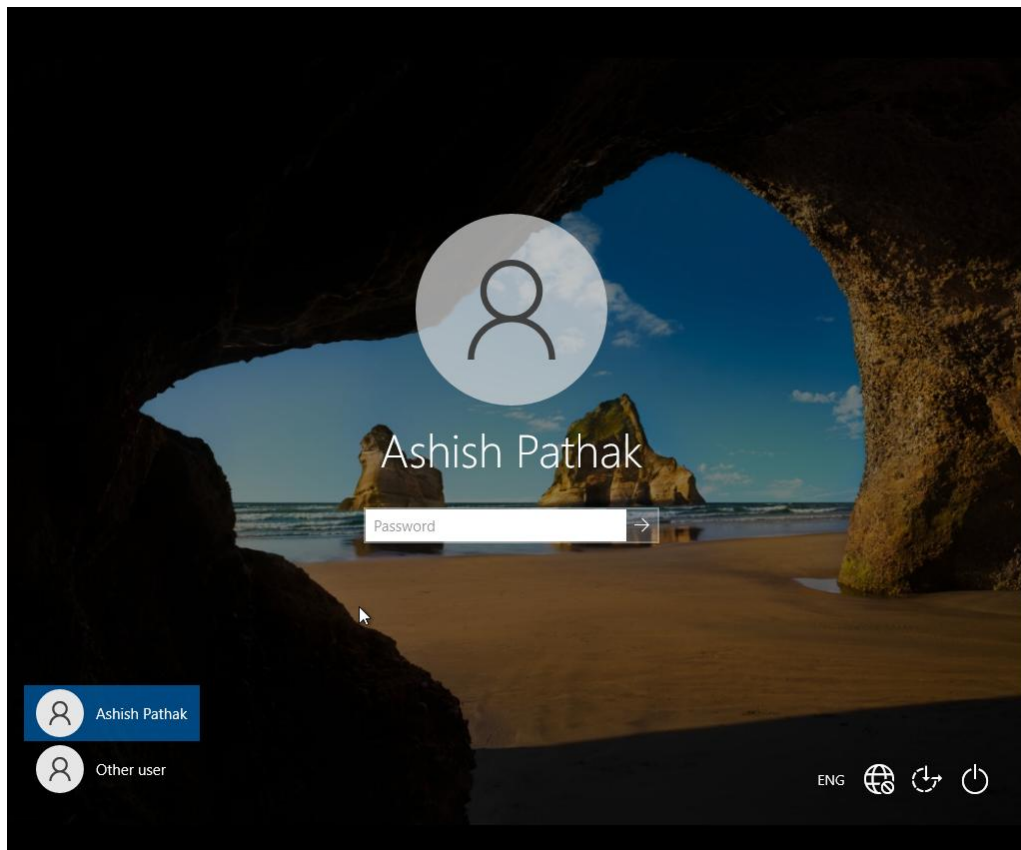




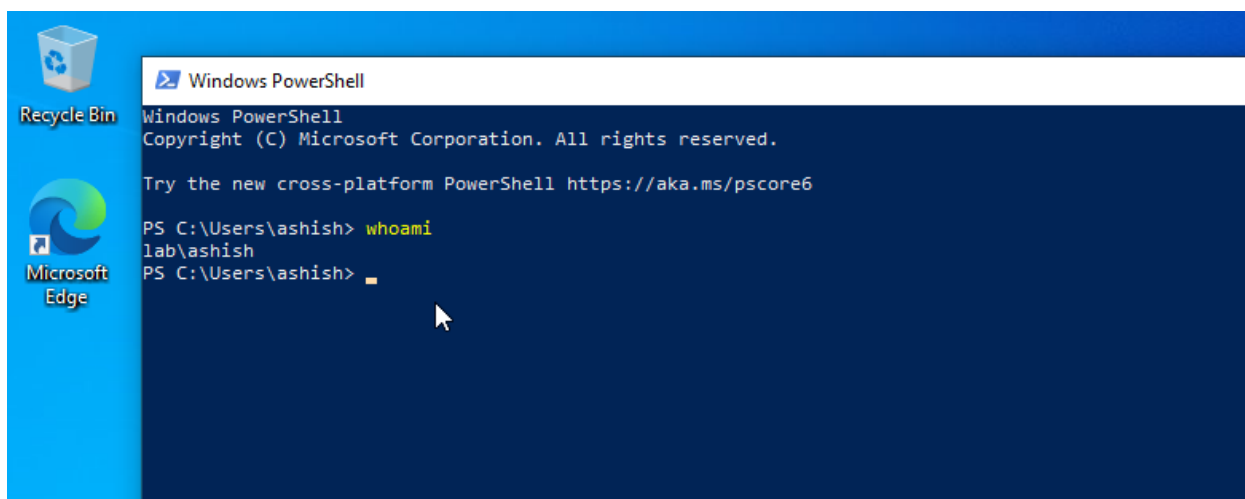
**Figure 9.3: Domain user accounts within the IT Organizational Unit**



**Figure 9.4: Security group membership configuration.**



*Figure 9.6: Domain user authentication using Active Directory credentials.*



*Figure 9.7: Verification of successful domain authentication.*

# DNS Configuration

The Domain Name System (DNS) was configured on the Windows Server 2022 Domain Controller to provide name resolution services for the enterprise network. DNS enables clients to locate domain controllers, resolve hostnames to IP addresses, and support Active Directory authentication and other network services.

## Overview

The DNS Server role was installed and configured on the Domain Controller (DC01) during the deployment of Active Directory Domain Services (AD DS). A Forward Lookup Zone named **lab.local** was automatically created as part of the Active Directory installation, enabling centralized name resolution for all devices within the domain. To improve DNS functionality and support network administration, additional Host (A), Alias (CNAME), and Pointer (PTR) records were created. These records allow systems within the enterprise network to communicate using hostnames instead of IP addresses.

---

## DNS Server Information

| Parameter       | Value                       |
|-----------------|-----------------------------|
| DNS Server      | DC01                        |
| IP Address      | 192.168.10.10               |
| Domain Name     | lab.local                   |
| DNS Zone Type   | Active Directory Integrated |
| Dynamic Updates | Secure Only                 |

---

## Forward Lookup Zone

The Forward Lookup Zone **lab.local** stores mappings between hostnames and IPv4 addresses. During the project, the following DNS records were configured.

### Host (A) Records

| Host Name  | IP Address    | Purpose                  |
|------------|---------------|--------------------------|
| dc01       | 192.168.10.10 | Domain Controller        |
| fileserver | 192.168.10.50 | File Server (Simulation) |

---

## Alias (CNAME) Record

Alias Points To

fs    fileserver.lab.local

The CNAME record allows users to access the file server using a simplified alias instead of the full hostname.

---

## Reverse Lookup Zone

A Reverse Lookup Zone for the **192.168.10.0/24** network was created to enable reverse DNS resolution.

The following PTR record was successfully generated.

| IP Address | Host Name |
|------------|-----------|
|------------|-----------|

192.168.10.10 dc01.lab.local

Reverse DNS resolution allows an IP address to be translated back into its corresponding hostname, which is useful for troubleshooting and network diagnostics.

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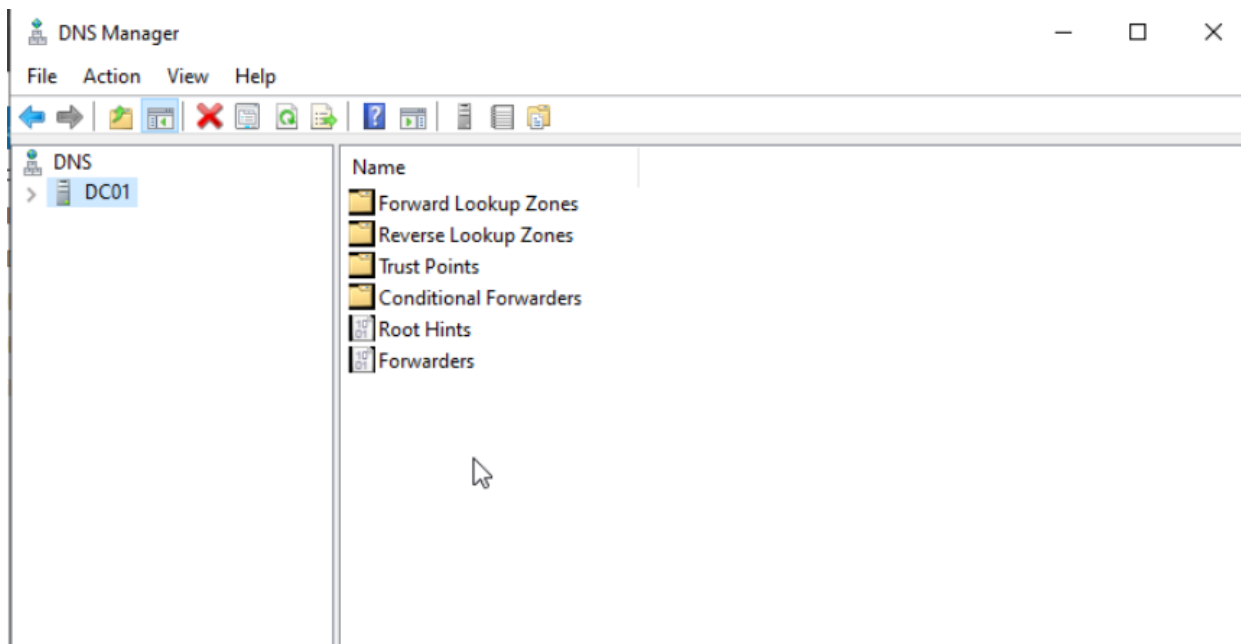
## DNS Testing

The DNS configuration was validated using Windows Command Prompt.

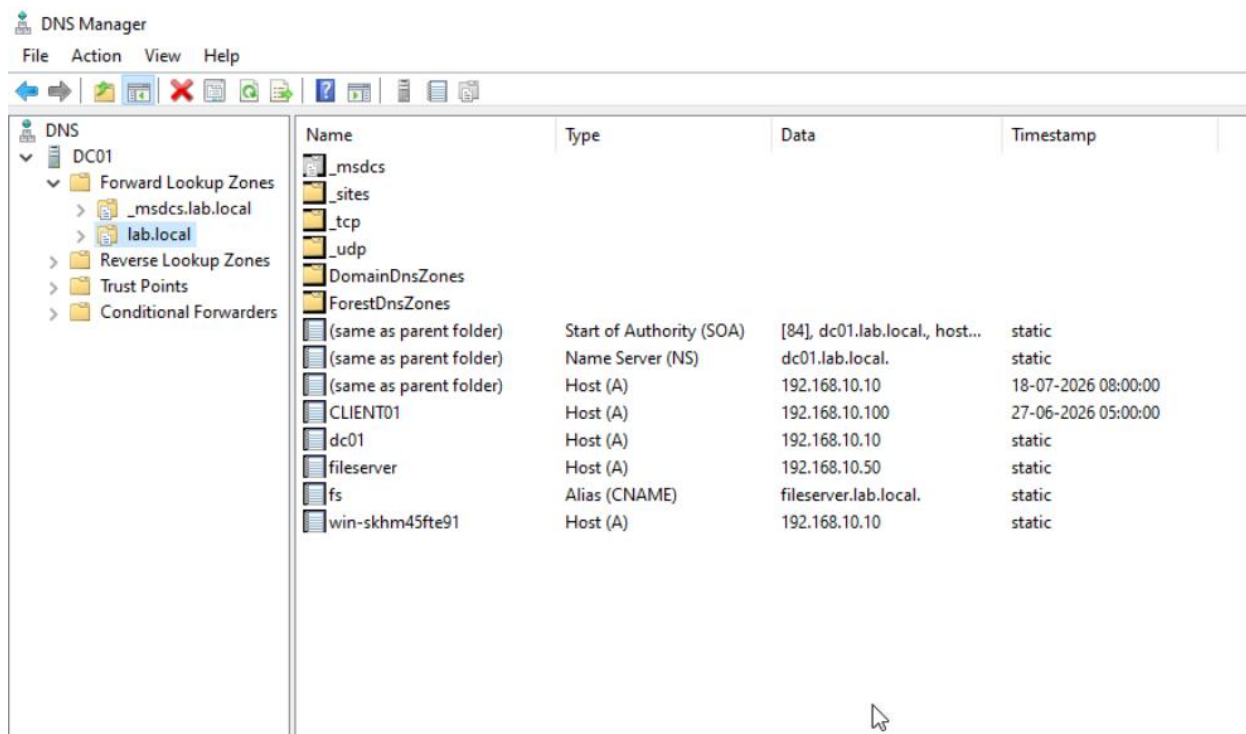
The following tests were performed successfully:

- Hostname resolution using **ping**
- Forward lookup using **nslookup**
- Reverse lookup using **nslookup**
- Domain name resolution for **lab.local**
- Verification of A, CNAME, and PTR records

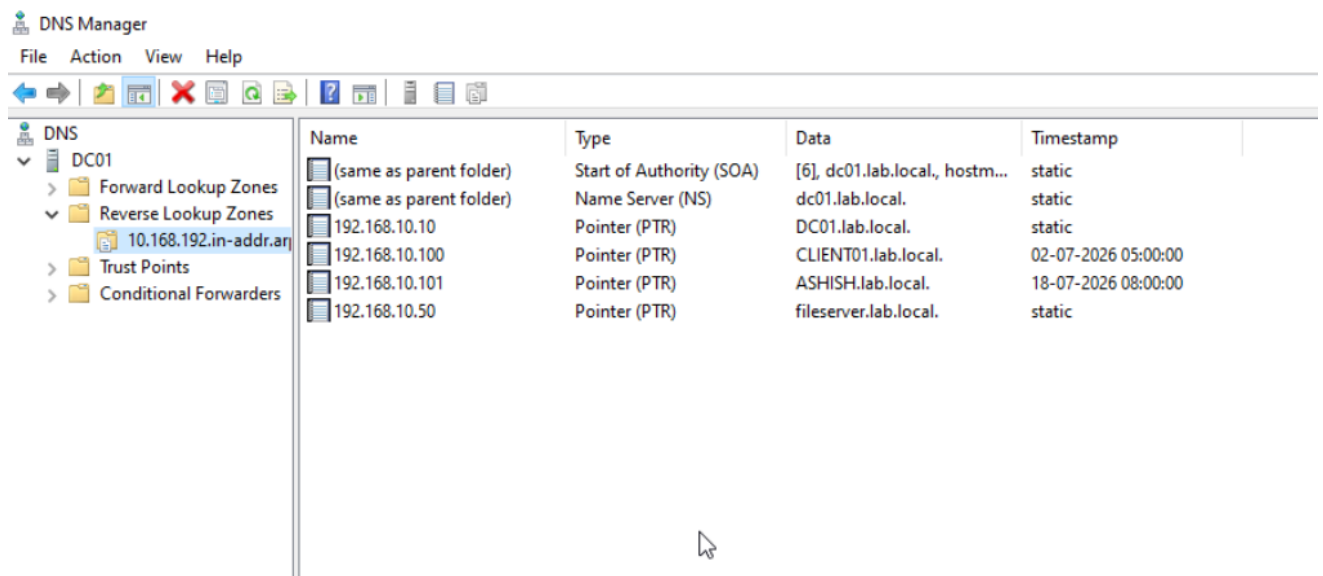
The successful completion of these tests confirmed that DNS services were functioning correctly and supporting Active Directory operations.



*Figure 10.1: DNS Manager showing the configured DNS zones.*



**Figure 10.2: Forward Lookup Zone (lab.local) with configured DNS records.**



**Figure 10.3: Reverse Lookup Zone containing the PTR record for the Domain Controller.**

```
C:\Users\ashish>nslookup lab.local
Server:  DC01.lab.local
Address: 192.168.10.10

Name:    lab.local
Address: 192.168.10.10
```

*Figure 10.4: Verification of DNS resolution for the lab.local domain.*

```
C:\Users\ashish>nslookup dc01.lab.local
Server:  DC01.lab.local
Address: 192.168.10.10

Name:    dc01.lab.local
Address: 192.168.10.10
```

*Figure 10.5: Forward DNS lookup for the Domain Controller.*

```
C:\Users\ashish>nslookup 192.168.10.10
Server:  DC01.lab.local
Address: 192.168.10.10

Name:    DC01.lab.local
Address: 192.168.10.10
```

*Figure 10.6: Reverse DNS lookup using the PTR record.*

```
C:\Users\ashish>ping fileserver

Pinging fileserver.lab.local [192.168.10.50] with 32 bytes of data:
Reply from 192.168.10.100: Destination host unreachable.
Reply from 192.168.10.100: Destination host unreachable.
Reply from 192.168.10.100: Destination host unreachable.
Reply from 192.168.10.100: Destination host unreachable.

Ping statistics for 192.168.10.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

*Figure 10.7: Hostname resolution using DNS A and CNAME records.*

## **DHCP Configuration**

The Dynamic Host Configuration Protocol (DHCP) service was configured on the Windows Server 2022 Domain Controller (DC01) to automate IP address allocation and simplify network administration. DHCP eliminates the need for manual IP configuration by dynamically assigning network parameters to client devices joining the enterprise network.

### **Overview**

The DHCP Server role was installed and configured on the Domain Controller (DC01) to provide automatic IP address assignment to client computers within the enterprise network. After installation, the DHCP server was authorized in Active Directory to enable it to distribute IP addresses securely across the domain.

A DHCP scope named **LabScope** was created to define the IPv4 address range available for client devices. In addition to assigning IP addresses, the DHCP server provides clients with the subnet mask, default gateway, DNS server address, and domain name required for network communication.

---

### **DHCP Server Information**

| Parameter         | Value                           |
|-------------------|---------------------------------|
| DHCP Server       | DC01                            |
| Server IP Address | 192.168.10.10                   |
| Scope Name        | LabScope                        |
| Network Address   | 192.168.10.0/24                 |
| Subnet Mask       | 255.255.255.0                   |
| Address Pool      | 192.168.10.100 – 192.168.10.200 |
| Lease Duration    | Default                         |
| DNS Server        | 192.168.10.10                   |
| Domain Name       | lab.local                       |

---

### **DHCP Scope Configuration**

The DHCP scope was configured to distribute IP addresses automatically to domain-joined client computers. The address range was selected to avoid conflicts with statically assigned infrastructure devices such as the Domain Controller.

The configured DHCP scope includes:

- Scope Name: LabScope
- Start IP Address: 192.168.10.100
- End IP Address: 192.168.10.200
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1
- Preferred DNS Server: 192.168.10.10
- DNS Domain Name: lab.local

---

### **DHCP Authorization**

After installing the DHCP Server role, the server was authorized in Active Directory using domain administrator credentials. This authorization ensures that only trusted DHCP servers within the Active Directory environment can distribute IP addresses to client systems.

---

### **DHCP Lease Allocation**

After configuring the DHCP scope, CLIENT01 was set to obtain its network configuration automatically. Upon renewal, the client successfully received an IPv4 address from the configured DHCP address pool along with the appropriate subnet mask, default gateway, DNS server, and domain information.

---

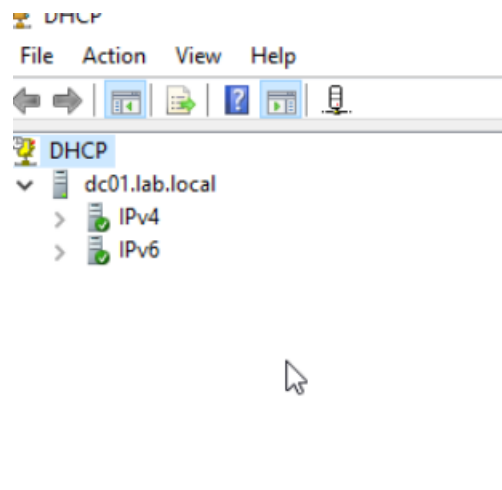
### **DHCP Validation**

The DHCP configuration was verified using the following methods:

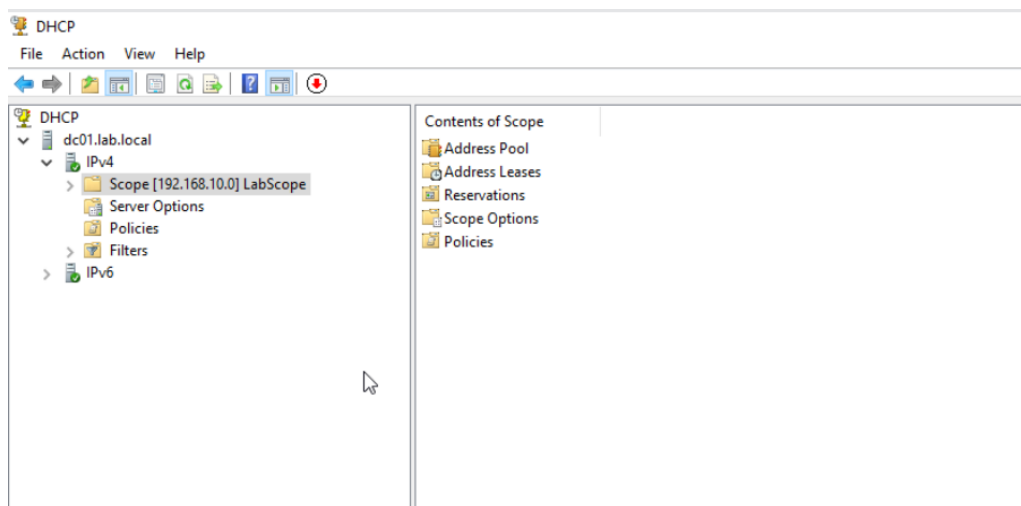
- Verification of the configured DHCP scope
- Inspection of the configured address pool
- Review of active DHCP leases
- Client network configuration using **ipconfig /all**
- DHCP lease renewal using **ipconfig /release** and **ipconfig /renew**

The successful completion of these validation steps confirmed that the DHCP server was functioning correctly and providing automatic network configuration to client devices.

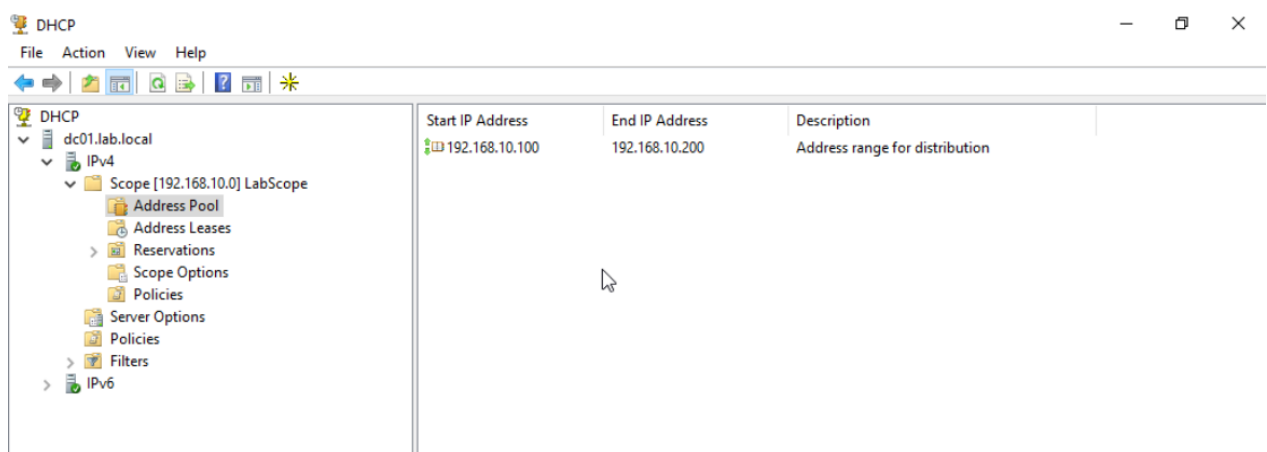




**Figure 11.1: DHCP Manager displaying the configured IPv4 scope.**



**Figure 11.2: Configured DHCP Scope (LabScope).**



**Figure 11.3: DHCP Address Pool Configuration.**

| DHCP                  |                  |                     |      |              |             |                           |                      |        |
|-----------------------|------------------|---------------------|------|--------------|-------------|---------------------------|----------------------|--------|
| File Action View Help |                  |                     |      |              |             |                           |                      |        |
| Client IP Address     | Name             | Lease Expiration    | Type | Unique ID    | Description | Network Access Protection | Probation Expiration | Filter |
| 192.168.10.101        | ASHISH.lab.local | 25-07-2026 21:49:26 | DHCP | 005056c00... |             | Full Access               | N/A                  | None   |

**Figure 11.4: Active DHCP lease assigned to CLIENT01.**

| DHCP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                     |          |               |             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------|---------------|-------------|
| File Action View Help                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                     |          |               |             |
| DHCP <ul style="list-style-type: none"> <li>dc01.lab.local               <ul style="list-style-type: none"> <li>IPv4                   <ul style="list-style-type: none"> <li>Scope [192.168.10.0] LabS                       <ul style="list-style-type: none"> <li>Address Pool</li> <li>Address Leases</li> <li>Reservations</li> <li>Scope Options</li> <li>Policies</li> <li>Server Options</li> <li>Policies</li> <li>Filters</li> <li>IPv6</li> </ul> </li> </ul> </li> </ul> </li> </ul> | Option Name         | Vendor   | Value         | Policy Name |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 003 Router          | Standard | 192.168.10.1  | None        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 006 DNS Servers     | Standard | 192.168.10.10 | None        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 015 DNS Domain Name | Standard | lab.local     | None        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                     |          |               |             |

**Figure 11.5: DHCP Scope Options distributed to client computers.**

cmd Command Prompt

```
C:\Users\ashish>ipconfig /all
```

Windows IP Configuration

```
Host Name . . . . . : CLIENT01
Primary Dns Suffix . . . . . : lab.local
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : lab.local
```

Ethernet adapter Ethernet0:

```
Connection-specific DNS Suffix . : lab.local
Description . . . . . : Intel(R) 82574L Gigabit Network Connection
Physical Address. . . . . : 00-0C-29-23-EF-0B
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::fe3c:c414:2dc8:f9f5%11(Preferred)
IPv4 Address. . . . . : 192.168.10.100(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 18 July 2026 20:52:41
Lease Expires . . . . . : 26 July 2026 20:52:40
Default Gateway . . . . . : 192.168.10.1
DHCP Server . . . . . : 192.168.10.10
DHCPv6 IAID . . . . . : 100666409
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-AA-57-EF-00-0C-29-23-EF-0B
DNS Servers . . . . . : 192.168.10.10
NetBIOS over Tcpip. . . . . : Enabled
```

```
C:\Users\ashish>
```

*Figure 11.6: CLIENT01 receiving network configuration from the DHCP server.*

# Cisco Network Configuration (VLANs, Trunking, Router-on-a-Stick)

Cisco Packet Tracer was used to design and configure the network infrastructure supporting the Windows Server environment. The network was segmented using Virtual Local Area Networks (VLANs) to separate server and client traffic. A Cisco router was configured using the Router-on-a-Stick technique to provide inter-VLAN routing, allowing secure communication between different network segments while maintaining logical isolation.

## Overview

The enterprise network topology was implemented using a Cisco router and a Cisco Layer-2 switch in Cisco Packet Tracer. VLAN segmentation was used to logically separate server and client devices into different broadcast domains, improving network organization, scalability, and security. A trunk link was established between the router and switch using IEEE 802.1Q encapsulation. Router-on-a-Stick was implemented by creating multiple router sub-interfaces, each acting as the default gateway for a specific VLAN.

---

## Network Topology

The network consists of one Cisco router (R1), one Cisco 2960 switch (SW1), one Windows Server (DC01), and one client workstation (CLIENT01). The router provides inter-VLAN routing while the switch manages Layer-2 connectivity and VLAN segmentation.

---

## VLAN Configuration

Two VLANs were created to separate enterprise resources.

| VLAN ID | VLAN Name | Purpose                       | Network         |
|---------|-----------|-------------------------------|-----------------|
| 10      | SERVERS   | Windows Server Infrastructure | 192.168.10.0/24 |
| 20      | CLIENTS   | Client Computers              | 192.168.20.0/24 |

Server systems were connected to VLAN 10, while client systems were connected to VLAN 20.

---

## Access Port Configuration

The switch access ports were configured to assign end devices to their respective VLANs.

| Switch Port     | Connected Device | VLAN    |
|-----------------|------------------|---------|
| FastEthernet0/2 | DC01             | VLAN 10 |
| FastEthernet0/3 | CLIENT01         | VLAN 20 |

---

## Trunk Configuration

A trunk link was configured between the Cisco router and the Cisco switch using IEEE 802.1Q trunking. The trunk port carries traffic for both VLAN 10 and VLAN 20 over a single physical connection.

| Device     | Interface          | Mode  |
|------------|--------------------|-------|
| Router R1  | GigabitEthernet0/0 | Trunk |
| Switch SW1 | GigabitEthernet0/1 | Trunk |

---

## Router-on-a-Stick Configuration

Inter-VLAN routing was implemented using Router-on-a-Stick. Separate logical sub-interfaces were created on the router for each VLAN.

| Sub-interface         | VLAN | IP Address   |
|-----------------------|------|--------------|
| GigabitEthernet0/0.10 | 10   | 192.168.10.1 |
| GigabitEthernet0/0.20 | 20   | 192.168.20.1 |

Each sub-interface serves as the default gateway for devices within its VLAN.

---

## Inter-VLAN Communication

Router-on-a-Stick enables communication between devices located in different VLANs while maintaining logical network separation.

For example:

- CLIENT01 can communicate with the Domain Controller (DC01).
  - CLIENT01 can access DNS and DHCP services hosted on DC01.
  - Domain authentication functions correctly across VLAN boundaries.
- 

## Verification Commands

The following Cisco IOS commands were used to verify the network configuration.

### Switch Verification

- show vlan brief
- show interfaces trunk
- show running-config

### Router Verification

- show ip interface brief
- show ip route
- show running-config

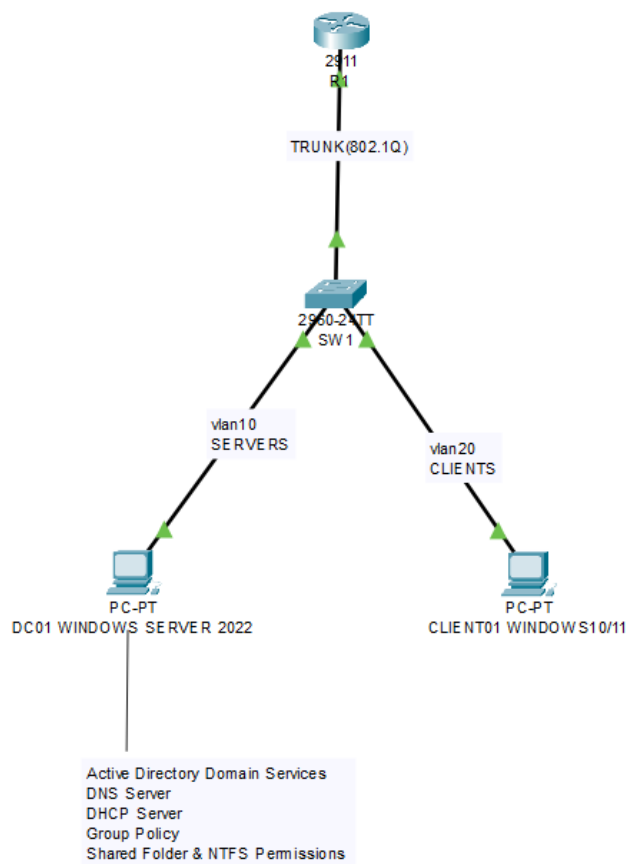
The successful output of these commands confirmed that VLANs, trunking, router sub-interfaces, and routing were configured correctly.

---

## Result

The Cisco network infrastructure successfully provided:

- VLAN-based network segmentation
- IEEE 802.1Q trunking
- Inter-VLAN routing using Router-on-a-Stick
- Connectivity between server and client networks
- Support for Windows Server services such as Active Directory, DNS, and DHCP
- A scalable enterprise LAN architecture following industry-standard networking practices



**Figure 12.1: Enterprise network topology designed in Cisco Packet Tracer.**

```

Switch>
Switch>en
Switch#show vlan brief

```

| VLAN | Name               | Status | Ports                                                                                                                                                                                     |
|------|--------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | default            | active | Fa0/1, Fa0/4, Fa0/5, Fa0/6<br>Fa0/7, Fa0/8, Fa0/9, Fa0/10<br>Fa0/11, Fa0/12, Fa0/13, Fa0/14<br>Fa0/15, Fa0/16, Fa0/17, Fa0/18<br>Fa0/19, Fa0/20, Fa0/21, Fa0/22<br>Fa0/23, Fa0/24, Gig0/2 |
| 10   | SERVERS            | active | Fa0/2                                                                                                                                                                                     |
| 20   | CLIEENTS           | active | Fa0/3                                                                                                                                                                                     |
| 1002 | fddi-default       | active |                                                                                                                                                                                           |
| 1003 | token-ring-default | active |                                                                                                                                                                                           |
| 1004 | fddinet-default    | active |                                                                                                                                                                                           |
| 1005 | trnet-default      | active |                                                                                                                                                                                           |

**Figure 12.2: VLAN configuration on the Cisco switch.**

```
Switch#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Gig0/1    1-1005

Port      Vlans allowed and active in management domain
Gig0/1    1,10,20

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    1,10,20

Switch#
```

---

**Figure 12.3: IEEE 802.1Q trunk configuration.**

```
R1#show ip in
R1#show ip interface bri
R1#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0    unassigned      YES unset   up          up
GigabitEthernet0/0.10 192.168.10.1    YES manual   up          up
GigabitEthernet0/0.20 192.168.20.1    YES manual   up          up
GigabitEthernet0/1    unassigned      YES unset   administratively down down
GigabitEthernet0/2    unassigned      YES unset   administratively down down
Vlan1            unassigned      YES unset   administratively down down
R1#
```

**Figure 12.4: Router subinterface configuration for Router-on-a-Stick.**

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
    192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0.20

R1#
```

**Figure 12.5: Router routing table.**

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time=2ms TTL=255
Reply from 192.168.20.1: bytes=32 time=1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

*Figure 12.6: Connectivity test between CLIENT01 and its default gateway.*

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time=11ms TTL=127

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 2ms
```

*Figure 12.7: Successful communication between CLIENT01 and the Windows Server across VLANs.*





```
interface Vlan1
  no ip address
  shutdown
  !
ip classless
!
ip flow-export version 9
!
!
access-list 100 deny tcp 192.168.20.0 0.0.0.255 any eq telnet
access-list 100 permit ip any any
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
  login local
  transport input ssh
!
!
!
end
```

R1#

---

*Figure 12.8: Router running configuration.*

# **Group Policy Configuration**

Group Policy Objects (GPOs) were implemented to centrally manage user and computer settings within the Active Directory domain. By applying policies through Active Directory, administrative tasks can be automated, security can be enhanced, and consistent configurations can be enforced across all domain-joined computers.

## **Overview**

Group Policy Objects (GPOs) were configured on the Windows Server 2022 Domain Controller to centrally manage user and computer configurations within the **lab.local** domain. The implemented policies demonstrate how enterprise administrators can enforce security settings, restrict user access, and automate administrative tasks without requiring manual configuration on individual client systems. The Group Policy Management Console (GPMC) was used to create, edit, and link policies to the appropriate Organizational Units (OUs) within Active Directory.

---

## **Group Policy Management**

The Group Policy Management Console was used to create and manage domain policies. After creating each policy, it was linked to the appropriate Organizational Unit to ensure that only the intended users or computers received the configuration.

---

## **Implemented Group Policies**

### **Control Panel Restriction**

A User Configuration policy was created to prevent users from accessing the Windows Control Panel and Settings application.

#### **Purpose:**

- Prevent unauthorized system configuration changes.
- Improve endpoint security.
- Enforce centralized administrative control.

The policy was successfully applied, and domain users were unable to access the Control Panel after logging into the domain.

---

## **Network Drive Mapping**

A Group Policy Preference was configured to automatically map the shared folder **\\DC01\IT\_Share** as drive **Z:** for users belonging to the appropriate Active Directory security group.

#### **Benefits:**

- Automatic resource availability after user logon.
- No manual drive mapping required.
- Centralized deployment of shared resources.

---

## **Group Policy Update**

After creating or modifying Group Policies, the client computer was updated using the following command:

`gpupdate /force`

The command successfully refreshed both computer and user policies without requiring manual configuration.

---

### **Policy Verification**

The successful application of Group Policies was verified using the following methods:

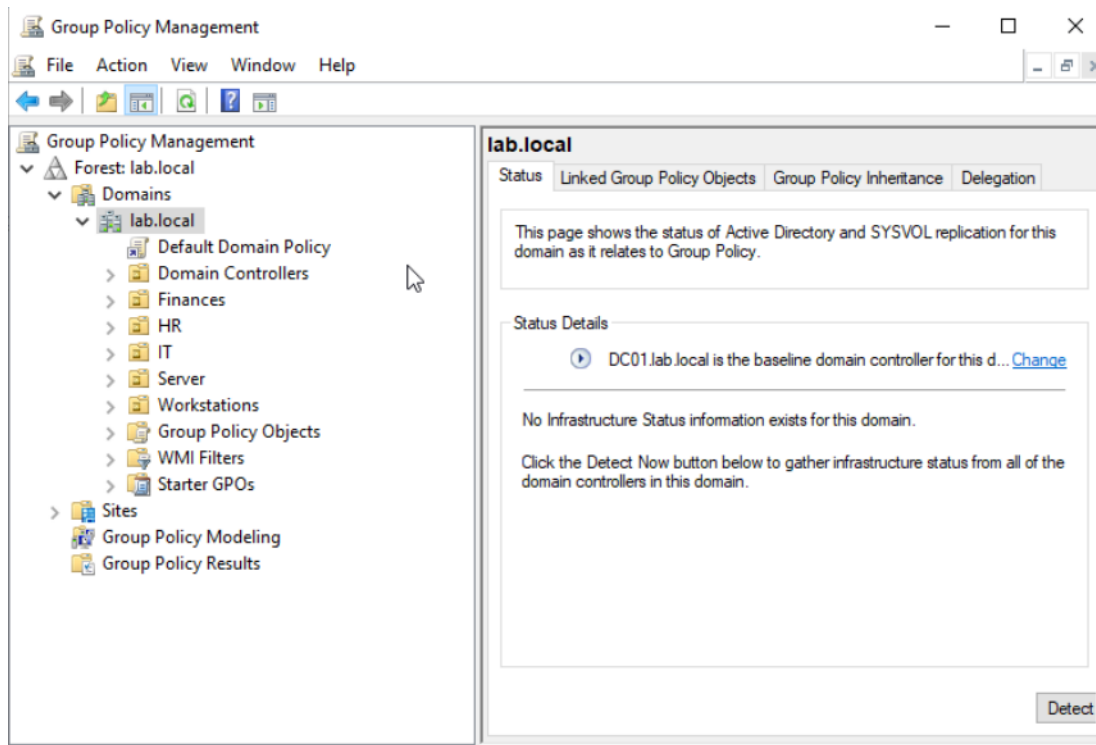
- Attempting to open the Control Panel from CLIENT01.
  - Confirming the automatic mapping of the Z: network drive.
  - Running **gpresult /r** to verify applied Group Policies.
  - Logging in with a domain user account and validating policy enforcement.
- 

### **Benefits of Group Policy**

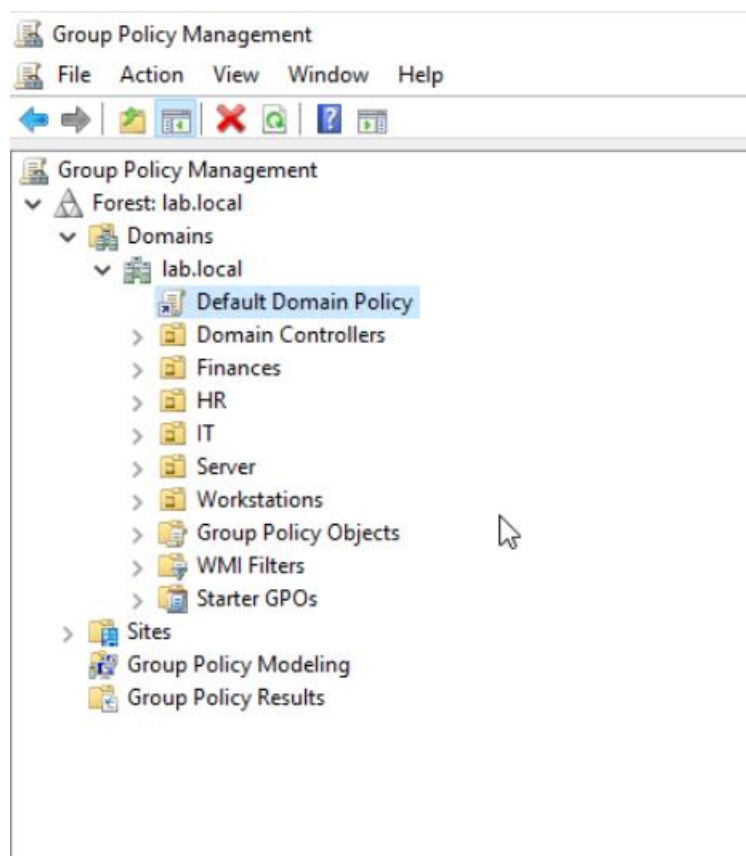
The implemented Group Policies provide several enterprise administration advantages:

- Centralized management of user and computer settings.
- Improved endpoint security.
- Reduced administrative effort.
- Consistent configuration across domain-joined systems.
- Automated deployment of network resources.

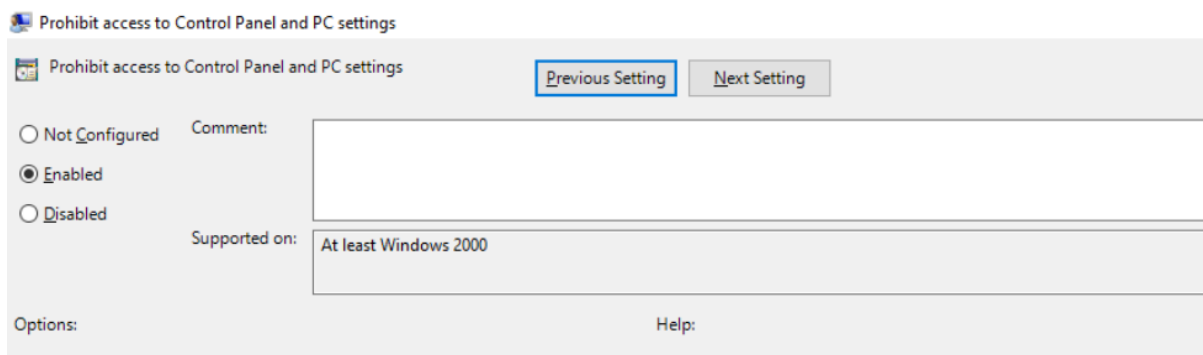
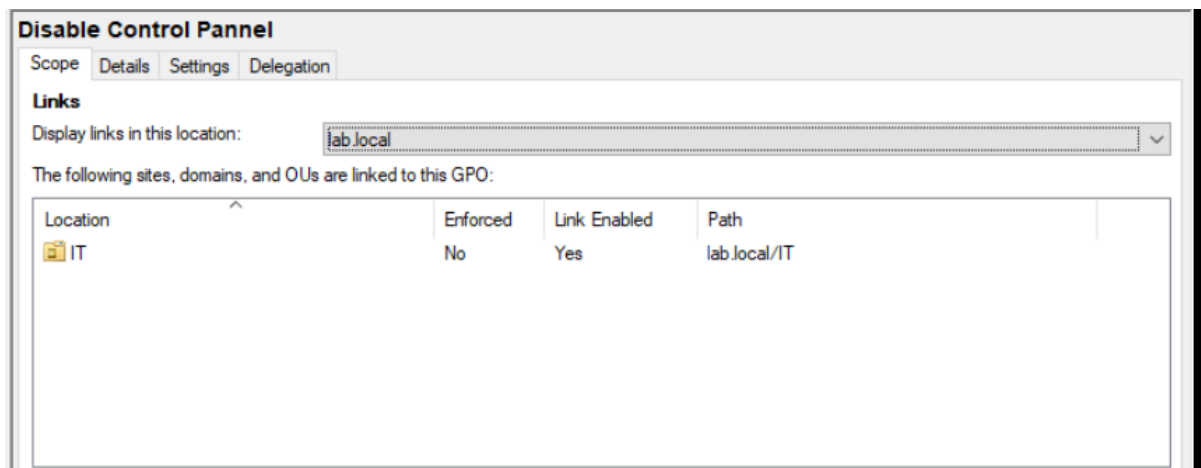
The successful implementation of Group Policy demonstrates the ability to centrally administer enterprise workstations using Windows Server 2022 and Active Directory.



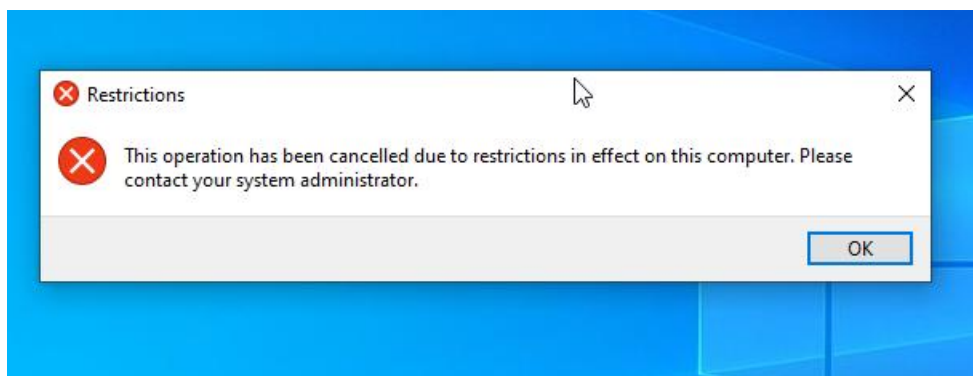
**Figure 13.1: Group Policy Management Console.**



**Figure 13.2: Group Policy linked to the Organizational Unit.**



**Figure 13.3: Configuration of the Control Panel restriction policy.**



**Figure 13.4: Control Panel access blocked by Group Policy.**

```
C:\Users\ashish>gpupdate /force
Updating policy...

Computer Policy update has completed successfully.
User Policy update has completed successfully.

C:\Users\ashish>
```

*Figure 13.5: Updating Group Policy on the client computer.*

```
Command Prompt
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ashish>gpresult /r

Microsoft (R) Windows (R) Operating System Group Policy Result tool v2.0
© Microsoft Corporation. All rights reserved.

Created on [19-07-2026 at 20:17:09]

RSOP data for LAB\ashish on CLIENT01 : Logging Mode
-----
OS Configuration:      Member Workstation
OS Version:            10.0.19045
Site Name:              N/A
Roaming Profile:        N/A
Local Profile:          C:\Users\ashish
Connected over a slow link?: No

USER SETTINGS
-----
Last time Group Policy was applied: 19-07-2026 at 20:16:30
Group Policy was applied from:      DC01.lab.local
Group Policy slow link threshold:   500 kbps
Domain Name:                        LAB
Domain Type:                        WindowsNT 4

Applied Group Policy Objects
-----
  Disable Control Panel
  Drive Mapping

The following GPOs were not applied because they were filtered out
-----
  Local Group Policy
  Filtering: Not Applied (Empty)

The user is a part of the following security groups
-----
  Everyone
```

```
USER SETTINGS
-----

Last time Group Policy was applied: 19-07-2026 at 20:16:30
Group Policy was applied from: DC01.lab.local
Group Policy slow link threshold: 500 kbps
Domain Name: LAB
Domain Type: WindowsNT 4

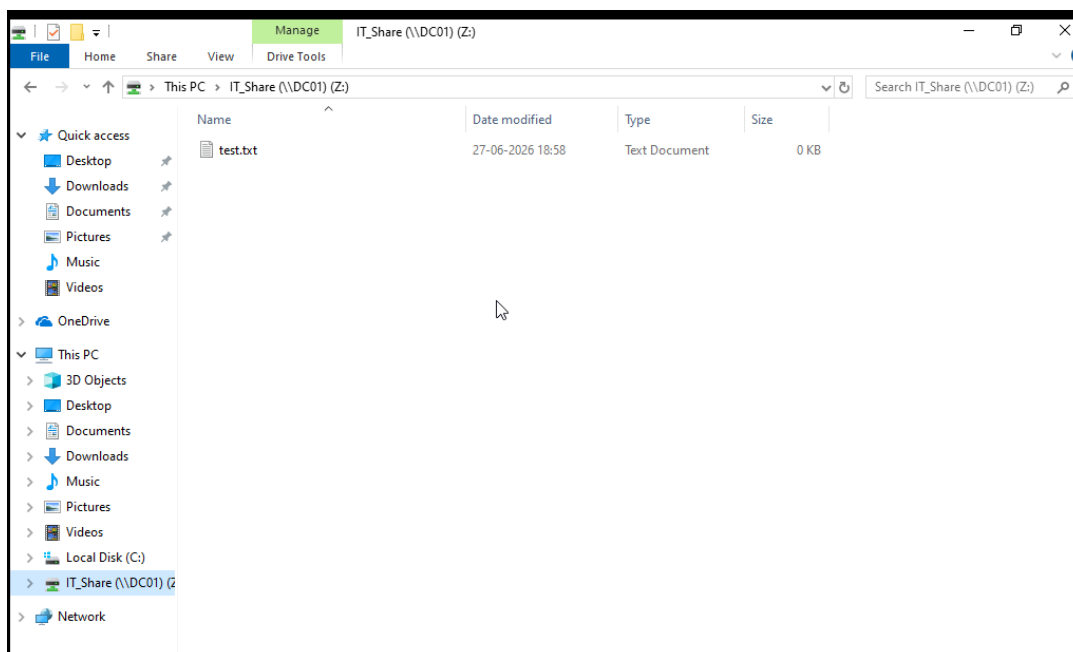
Applied Group Policy Objects
-----
    Disable Control Pannel
    Drive Mapping

The following GPOs were not applied because they were filtered out
-----
    Local Group Policy
    Filtering: Not Applied (Empty)

The user is a part of the following security groups
-----
    Everyone
    BUILTIN\Users
    NT AUTHORITY\INTERACTIVE
    CONSOLE LOGON
    NT AUTHORITY\Authenticated Users
    This Organization
    LOCAL
    Authentication authority asserted identity
    Medium Mandatory Level

C:\Users\ashish>
```

**Figure 13.6: Verification of applied Group Policies using gpresult**



**Figure 13.7: Automatically mapped network drive deployed through Group Policy.**



# Shared Folder & NTFS Permissions

Shared folders and NTFS permissions were configured to provide secure, role-based access to organizational resources. Access control was implemented using Active Directory security groups, allowing permissions to be managed centrally rather than assigning rights directly to individual users. This approach follows enterprise best practices by simplifying administration and improving security.

## Overview

A shared folder named **IT\_Share** was created on the Windows Server 2022 Domain Controller (DC01) to provide centralized file sharing for domain users. Access to the shared folder was controlled using both Share Permissions and NTFS Permissions, ensuring that only authorized users could access the resources.

Instead of assigning permissions directly to user accounts, Active Directory security groups were used to simplify permission management and improve scalability.

---

## Shared Folder Configuration

The shared folder was created on the Domain Controller and shared with the following settings.

| Parameter          | Value           |
|--------------------|-----------------|
| Shared Folder Name | IT_Share        |
| Server Name        | DC01            |
| Network Path       | \\DC01\IT_Share |
| Share Type         | SMB File Share  |

---

## Share Permissions

Share permissions determine what users can do when accessing the folder over the network.

The following permissions were configured:

Security Principal Permission

IT\_Users              Change, Read

Administrators      Full Control

The default **Everyone** group was removed to prevent unauthorized access.

---

## NTFS Permissions

NTFS permissions provide file system-level security and work together with Share Permissions to control access.

The following permissions were configured:

Security Principal Permission

IT\_Users              Modify

Administrators      Full Control

SYSTEM              Full Control

Inheritance was disabled where necessary to allow customized permissions for the shared folder.

---

### **Security Group Implementation**

A security group named **IT\_Users** was created within Active Directory.

Domain users requiring access to the shared folder were added to this group. Permissions were assigned to the group rather than individual user accounts, following the principle of role-based access control (RBAC).

This approach simplifies administration because future users only need to be added to the appropriate security group to inherit the required permissions.

---

### **Shared Folder Access**

The shared folder was successfully accessed from CLIENT01 using a domain user account.

A network drive (Z:) was automatically mapped through Group Policy to provide seamless access to the shared resource after user logon.

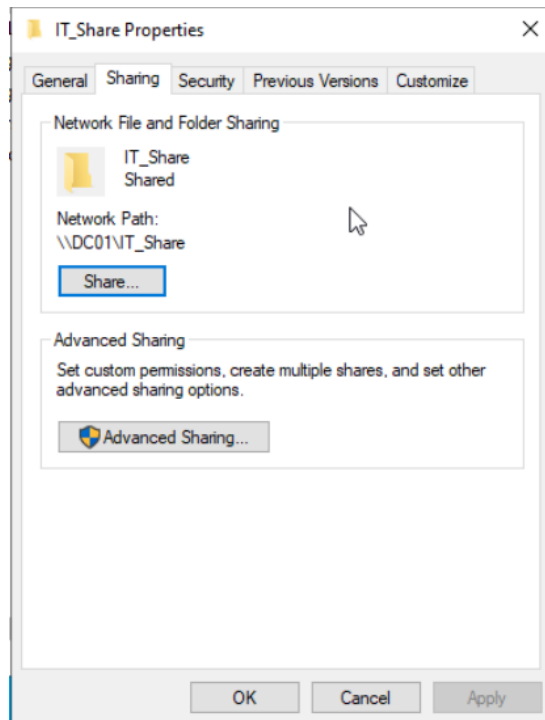
---

### **Verification**

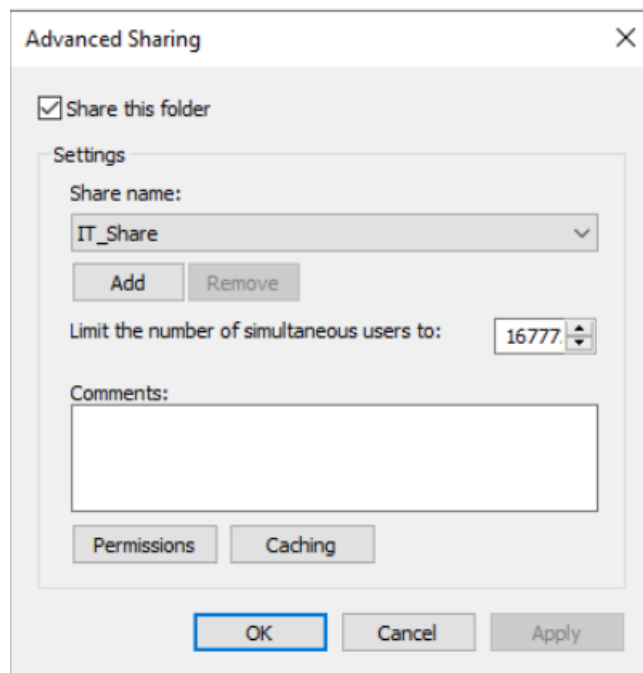
The shared folder configuration was validated using the following tests:

- Verification of Share Permissions.
- Verification of NTFS Permissions.
- Successful access to \\DC01\IT\_Share from CLIENT01.
- Successful automatic mapping of the Z: drive through Group Policy.
- Verification that authorized users could create, modify, and delete files within the shared folder.

The successful completion of these tests confirmed that the shared folder was securely configured and accessible only to authorized domain users.



*Figure 14.1: Shared folder configuration for IT\_Share.*



*Figure 14.2: Advanced Sharing configuration.*

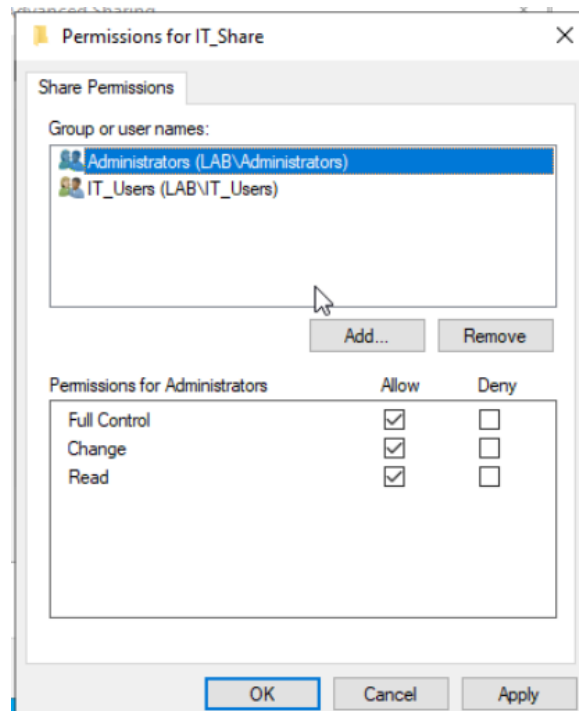


Figure 14.3: Share permissions configured for IT\_Share.

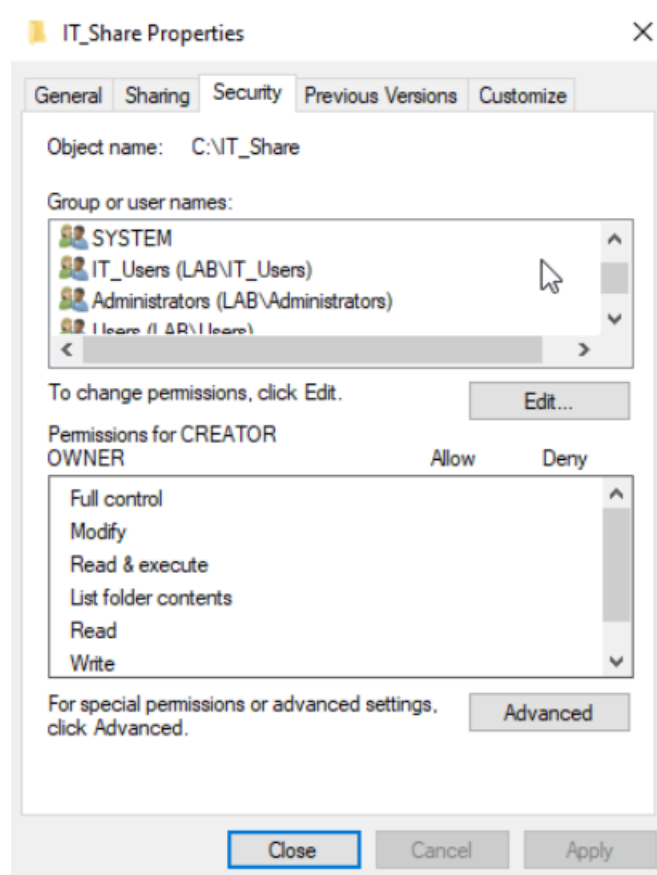
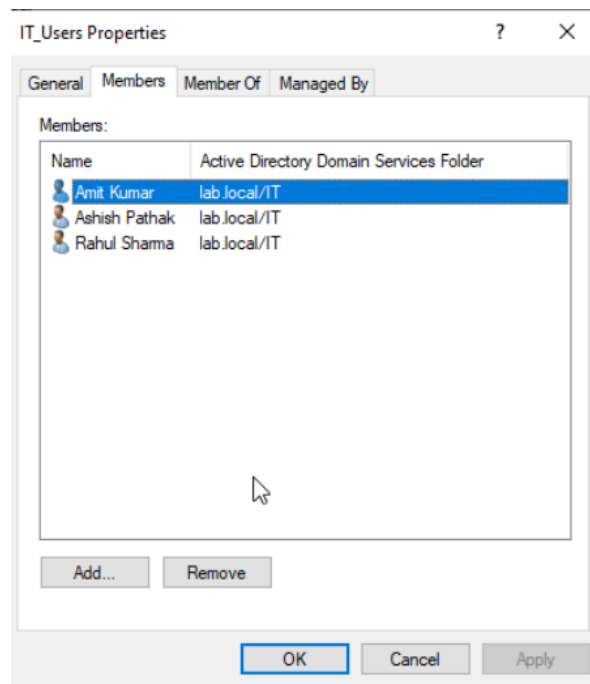
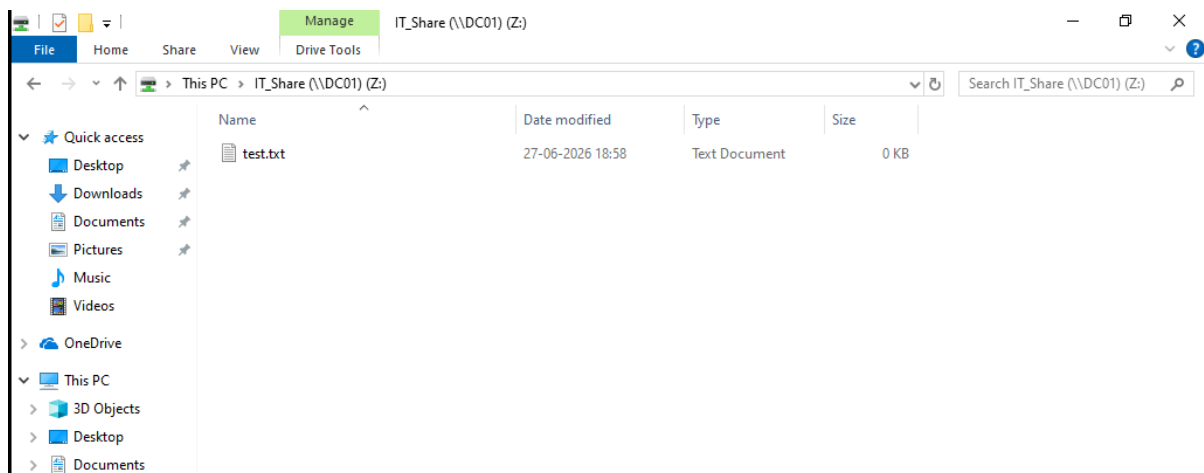


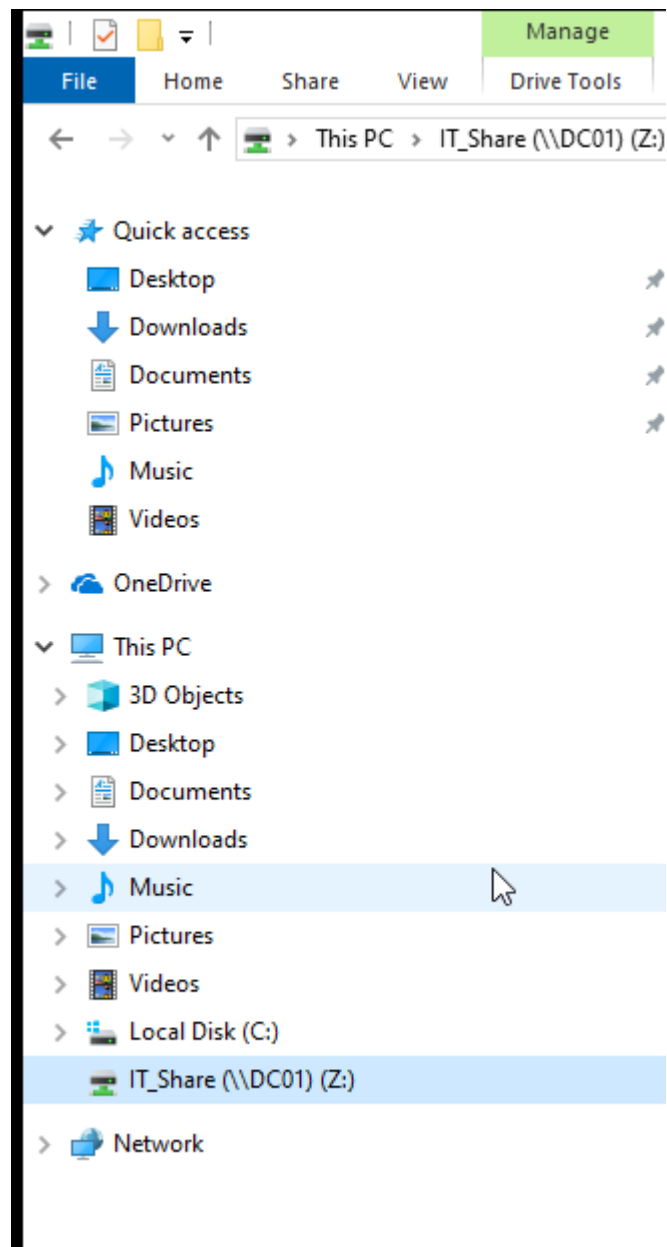
Figure 14.4: NTFS permissions assigned to the shared folder.



**Figure 14.5: Active Directory security group used for access control.**



**Figure 14.6: Successful access to the shared folder from a domain-joined client**



*Figure 14.7: Network drive automatically mapped using Group Policy.*

# Testing and Validation

After completing the configuration of the enterprise network infrastructure, a series of functional tests were performed to verify the correct operation of Windows Server services, Active Directory, DNS, DHCP, Group Policy, file sharing, and network connectivity. The objective of these tests was to ensure that all services operated together as an integrated enterprise environment.

## Overview

A comprehensive testing process was conducted to validate the functionality of the enterprise network environment. Each configured service was tested individually and then verified as part of the complete infrastructure to ensure reliable communication, centralized management, and secure resource access.

---

## Test Summary

| Test ID | Test Description                            | Expected Result                                | Status |
|---------|---------------------------------------------|------------------------------------------------|--------|
| TC-01   | Verify Client receives IP address from DHCP | Client receives valid IP address automatically | Passed |
| TC-02   | Verify Domain Join                          | CLIENT01 successfully joins lab.local          | Passed |
| TC-03   | Verify Domain Login                         | Domain user successfully authenticates         | Passed |
| TC-04   | Verify DNS Name Resolution                  | Hostnames resolve correctly                    | Passed |
| TC-05   | Verify Reverse DNS Lookup                   | PTR record resolves correctly                  | Passed |
| TC-06   | Verify Active Directory Users               | Domain users authenticate successfully         | Passed |
| TC-07   | Verify Group Policy Application             | Configured policies applied to client          | Passed |
| TC-08   | Verify Shared Folder Access                 | Authorized users access IT_Share               | Passed |
| TC-09   | Verify Network Drive Mapping                | Z: drive mapped automatically                  | Passed |
| TC-10   | Verify Network Connectivity                 | Successful communication between devices       | Passed |

---

## Test Case 1 – DHCP Verification

### Objective:

Verify that CLIENT01 receives its network configuration automatically.

### Procedure:

1. Configure CLIENT01 to obtain an IP address automatically.
2. Execute:

ipconfig /all

### Expected Result:

- DHCP Enabled = Yes
- IP Address assigned from LabScope
- DNS Server = 192.168.10.10
- Correct Default Gateway

**Actual Result:**

CLIENT01 successfully obtained network settings from the DHCP server.

**Status:** Passed

---

**Test Case 2 – Active Directory Authentication****Objective:**

Verify domain authentication.

**Procedure:**

Login using:

LAB\Ashish

**Expected Result:**

Successful authentication.

**Actual Result:**

User logged into the Active Directory domain successfully.

**Status:** Passed

---

**Test Case 3 – DNS Verification****Objective:**

Verify DNS functionality.

**Procedure:**

Execute:

nslookup dc01.lab.local

**Expected Result:**

Returns:

192.168.10.10

**Actual Result:**

DNS resolved successfully.

**Status:** Passed

---

**Test Case 4 – Reverse Lookup****Procedure:**

nslookup 192.168.10.10

**Expected Result:**

dc01.lab.local

**Actual Result:**

PTR record resolved correctly.

**Status:** Passed

---

**Test Case 5 – Group Policy Verification****Objective:**

Verify that Group Policy has been applied.

**Procedure:**

1. Execute:

gpupdate /force

2. Execute:

gpresult /r



3. Attempt to open Control Panel.

**Expected Result:**

Control Panel access blocked.

**Actual Result:**

Group Policy applied successfully.

**Status:** Passed

---

### **Test Case 6 – Shared Folder Access**

**Objective:**

Verify secure access to shared resources.

**Procedure:**

**Open:**

\\DC01\\IT\_Share

**Expected Result:**

Authorized users can access the folder.

**Actual Result:**

Folder opened successfully.

**Status:** Passed

---

### **Test Case 7 – Network Drive Mapping**

**Objective:**

Verify automatic drive mapping.

**Procedure:**

Log into CLIENT01.

**Expected Result:**

Drive Z: appears automatically.

**Actual Result:**

Drive mapped successfully.

**Status:** Passed

---

### **Test Case 8 – Network Connectivity**

**Objective:**

Verify communication between enterprise devices.

**Procedure:**

Execute:

ping dc01.lab.local

ping fileserver

**Expected Result:**

Replies received successfully.

**Actual Result:**

Connectivity verified.

**Status:** Passed

---

### **Overall Result**

All planned validation tests were completed successfully. The Windows Server 2022 infrastructure, Active Directory Domain Services, DNS, DHCP, Group Policy, shared folders,

and Cisco network components functioned together as an integrated enterprise network. The testing process confirmed that the environment met the project objectives and demonstrated the successful implementation of centralized authentication, automated IP address allocation, name resolution, policy enforcement, secure file sharing, and reliable network connectivity.

```
Command Prompt

C:\Users\ashish>ipconfig /all

Windows IP Configuration

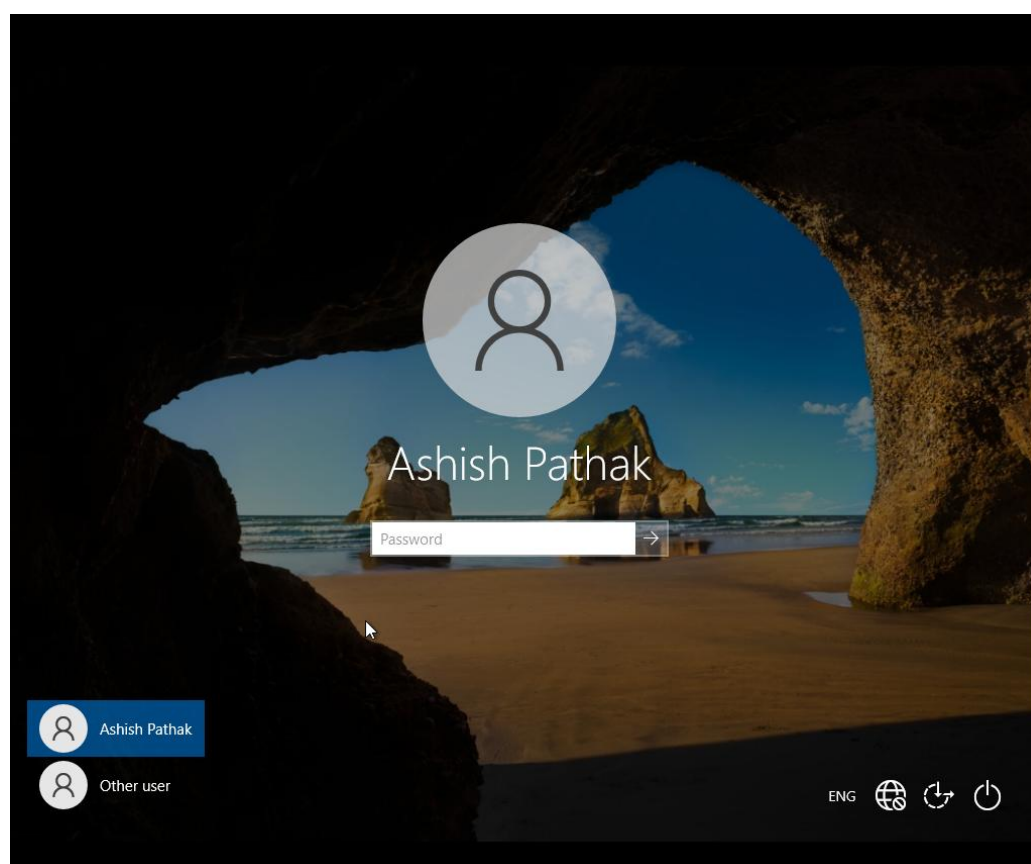
Host Name . . . . . : CLIENT01
Primary Dns Suffix . . . . . : lab.local
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : lab.local

Ethernet adapter Ethernet0:

Connection-specific DNS Suffix . : lab.local
Description . . . . . : Intel(R) 82574L Gigabit Network Connection
Physical Address. . . . . : 00-0C-29-23-EF-0B
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::fe3c:c414:2dc8:f9f5%11(Preferred)
IPv4 Address. . . . . : 192.168.10.100(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 18 July 2026 20:52:41
Lease Expires . . . . . : 26 July 2026 20:52:40
Default Gateway . . . . . : 192.168.10.1
DHCP Server . . . . . : 192.168.10.10
DHCPv6 IAID . . . . . : 100666409
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-AA-57-EF-00-0C-29-23-EF-0B
DNS Servers . . . . . : 192.168.10.10
NetBIOS over Tcpip. . . . . : Enabled

C:\Users\ashish>
```

**Figure 15.1** *ipconfig /all* on CLIENT01 showing DHCP configuration



**Figure 15.2** Domain login using LAB\Ashish

```
C:\Users\ashish>nslookup dc01.lab.local
Server:  DC01.lab.local
Address: 192.168.10.10

Name:    dc01.lab.local
Address: 192.168.10.10
```

*Figure 15.3 nslookup dc01.lab.local*

```
C:\Users\ashish>nslookup 192.168.10.10
Server:  DC01.lab.local
Address: 192.168.10.10

Name:    DC01.lab.local
Address: 192.168.10.10
```

*Figure 15.4 nslookup 192.168.10.10 (PTR lookup)*

```
Command Prompt
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ashish>gpresult /r

Microsoft (R) Windows (R) Operating System Group Policy Result tool v2.0
© Microsoft Corporation. All rights reserved.

Created on [19-07-2026 at 20:17:09]

RSOP data for LAB\ashish on CLIENT01 : Logging Mode
-----
OS Configuration:      Member Workstation
OS Version:            10.0.19045
Site Name:              N/A
Roaming Profile:        N/A
Local Profile:          C:\Users\ashish
Connected over a slow link?: No

USER SETTINGS
-----

Last time Group Policy was applied: 19-07-2026 at 20:16:30
Group Policy was applied from:      DC01.lab.local
Group Policy slow link threshold:   500 kbps
Domain Name:                        LAB
Domain Type:                        WindowsNT 4

Applied Group Policy Objects
-----
    Disable Control Panel
    Drive Mapping

The following GPOs were not applied because they were filtered out
-----
    Local Group Policy
        Filtering: Not Applied (Empty)

The user is a part of the following security groups
-----
    Everyone
```

```
USER SETTINGS
-----

Last time Group Policy was applied: 19-07-2026 at 20:16:30
Group Policy was applied from:      DC01.lab.local
Group Policy slow link threshold:  500 kbps
Domain Name:                       LAB
Domain Type:                       WindowsNT 4

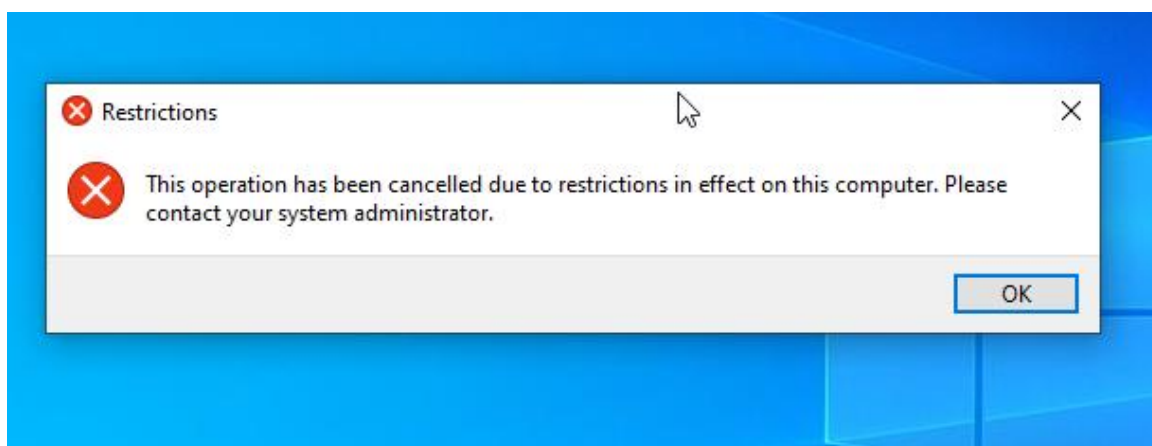
Applied Group Policy Objects
-----
    Disable Control Pannel
    Drive Mapping

The following GPOs were not applied because they were filtered out
-----
    Local Group Policy
    Filtering:  Not Applied (Empty)

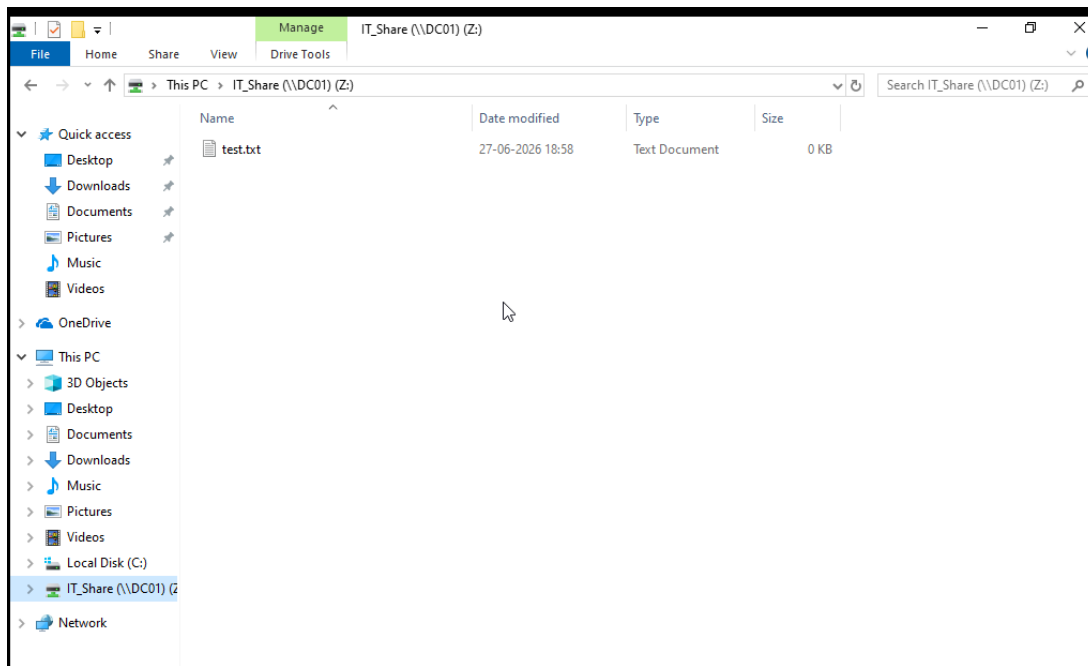
The user is a part of the following security groups
-----
    Everyone
    BUILTIN\Users
    NT AUTHORITY\INTERACTIVE
    CONSOLE LOGON
    NT AUTHORITY\Authenticated Users
    This Organization
    LOCAL
    Authentication authority asserted identity
    Medium Mandatory Level

C:\Users\ashish>
```

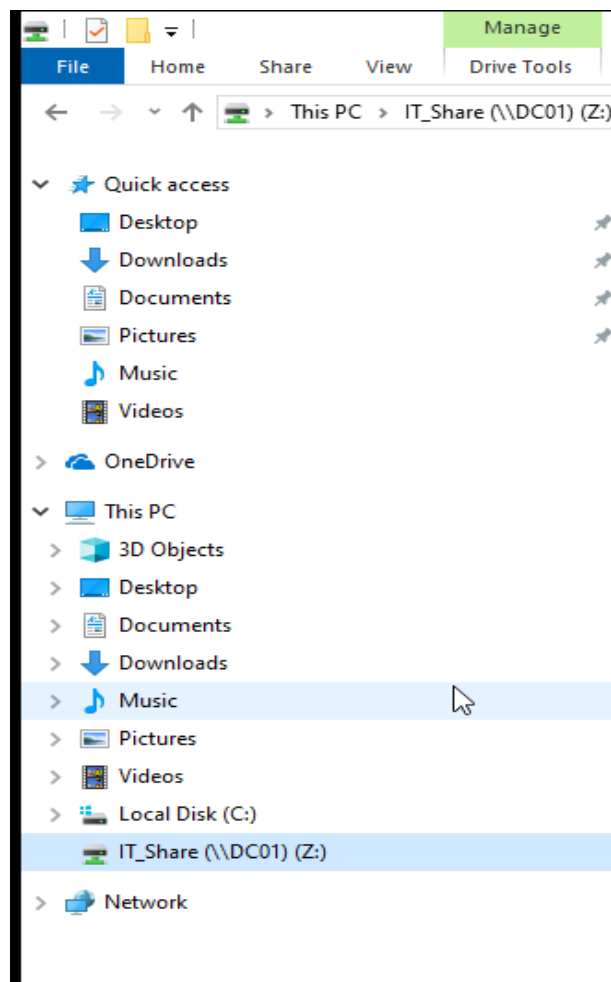
*Figure 15.5 gpresult /r showing applied GPOs*



*Figure 15.6 Control Panel restriction message*



**Figure 15.7** [\\DC01\IT\\_Share opened from CLIENT01](#)



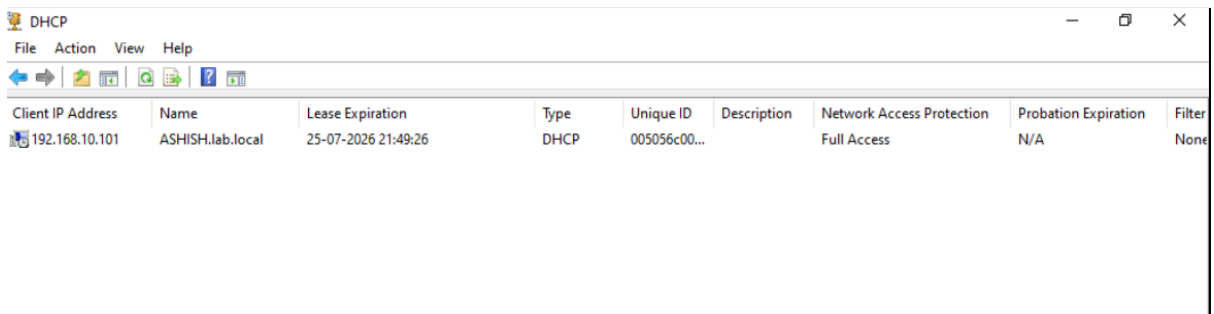
**Figure 15.8** Mapped Z: drive

```
C:\Users\ashish>ping fileserver

Pinging fileserver.lab.local [192.168.10.50] with 32 bytes of data:
Reply from 192.168.10.100: Destination host unreachable.
Reply from 192.168.10.100: Destination host unreachable.
Reply from 192.168.10.100: Destination host unreachable.
Reply from 192.168.10.100: Destination host unreachable.

Ping statistics for 192.168.10.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

*Figure 15.9 ping fileserver (or ping 192.168.10.50) showing successful replies*



The screenshot shows the DHCP console window with a single lease entry for CLIENT01. The table below represents the data shown in the console.

| Client IP Address | Name             | Lease Expiration    | Type | Unique ID    | Description | Network Access Protection | Probation Expiration | Filter |
|-------------------|------------------|---------------------|------|--------------|-------------|---------------------------|----------------------|--------|
| 192.168.10.101    | ASHISH.lab.local | 25-07-2026 21:49:26 | DHCP | 005056c00... |             | Full Access               | N/A                  | None   |

*Figure 15.10 DHCP Address Leases showing CLIENT01*

# Challenges Encountered and Troubleshooting

During the implementation of the enterprise network, several configuration and connectivity issues were encountered. Each issue was analyzed using standard troubleshooting techniques, and appropriate corrective actions were taken. Resolving these challenges improved the reliability and functionality of the overall network environment while strengthening practical troubleshooting skills.

## Overview

The deployment of the enterprise network required troubleshooting multiple issues related to Windows Server configuration, Active Directory, DNS, DHCP, Group Policy, and shared resources. The following table summarizes the key challenges encountered and the corresponding solutions implemented.

---

### Troubleshooting Summary

| Issue                                            | Cause                                                                     | Resolution                                                                                                                   |
|--------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Client could not join the domain                 | Incorrect DNS configuration                                               | Configured CLIENT01 to use DC01 (192.168.10.10) as the preferred DNS server and verified domain connectivity.                |
| Domain user account was disabled                 | User account was disabled in Active Directory                             | Enabled the user account in Active Directory Users and Computers and verified successful login.                              |
| DHCP client did not receive an IP address        | DHCP scope or client configuration issue                                  | Verified the DHCP scope, activated it, and configured CLIENT01 to obtain an IP address automatically.                        |
| DNS name resolution failed                       | Missing or incorrect DNS records                                          | Created and verified A, CNAME, and PTR records using DNS Manager and validated them with nslookup.                           |
| Reverse DNS lookup failed                        | Reverse Lookup Zone was not configured                                    | Created a Reverse Lookup Zone and verified PTR record resolution.                                                            |
| Internet DNS forwarding did not work             | The virtual environment used Host-Only networking without Internet access | Recognized that external DNS forwarding requires NAT or Bridged networking. Internal DNS functionality remained operational. |
| Group Policy was not applied                     | Policy update had not yet occurred                                        | Executed gpupdate /force and verified policy application using gpresult /r.                                                  |
| Control Panel restriction initially did not work | GPO was not linked or updated                                             | Linked the GPO to the correct Organizational Unit, refreshed policies, and confirmed successful enforcement.                 |
| Unable to modify NTFS permissions                | Permission inheritance was enabled                                        | Disabled inheritance on the shared folder and configured custom NTFS permissions.                                            |



Shared folder  
access failed

Incorrect Share or NTFS  
permissions

Assigned permissions to the IT\_Users  
security group and verified access from  
CLIENT01.

---

## **Troubleshooting Methodology**

The following tools and techniques were used throughout the project:

- Ping for connectivity testing.
  - ipconfig /all to verify network configuration.
  - nslookup for DNS verification.
  - gpupdate /force to refresh Group Policies.
  - gpresult /r to verify applied policies.
  - Active Directory Users and Computers for account management.
  - DNS Manager for record verification.
  - DHCP Manager for scope and lease validation.
  - Event Viewer for reviewing Windows Server events when required.
- 

## **Lessons Learned**

The troubleshooting process provided practical experience in diagnosing and resolving common enterprise networking issues. It reinforced the importance of proper DNS configuration for Active Directory, correct DHCP scope configuration, accurate permission management, and systematic verification after every configuration change. The project also highlighted the value of structured troubleshooting using built-in Windows administrative tools.

```
Command Prompt
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ashish>gpresult /r

Microsoft (R) Windows (R) Operating System Group Policy Result tool v2.0
© Microsoft Corporation. All rights reserved.

Created on [ 19-07-2026 at 20:17:09 ]

RSOP data for LAB\ashish on CLIENT01 : Logging Mode
-----

OS Configuration:      Member Workstation
OS Version:            10.0.19045
Site Name:              N/A
Roaming Profile:        N/A
Local Profile:          C:\Users\ashish
Connected over a slow link?: No

USER SETTINGS
-----

Last time Group Policy was applied: 19-07-2026 at 20:16:30
Group Policy was applied from:      DC01.lab.local
Group Policy slow link threshold:   500 kbps
Domain Name:                        LAB
Domain Type:                        WindowsNT 4

Applied Group Policy Objects
-----
Disable Control Panel
Drive Mapping

The following GPOs were not applied because they were filtered out
-----
Local Group Policy
Filtering: Not Applied (Empty)

The user is a part of the following security groups
-----
Everyone
```

```
USER SETTINGS
-----

Last time Group Policy was applied: 19-07-2026 at 20:16:30
Group Policy was applied from:      DC01.lab.local
Group Policy slow link threshold:   500 kbps
Domain Name:                        LAB
Domain Type:                        WindowsNT 4

Applied Group Policy Objects
-----
Disable Control Panel
Drive Mapping

The following GPOs were not applied because they were filtered out
-----
Local Group Policy
Filtering: Not Applied (Empty)

The user is a part of the following security groups
-----
Everyone
BUILTIN\Users
NT AUTHORITY\INTERACTIVE
CONSOLE LOGON
NT AUTHORITY\Authenticated Users
This Organization
LOCAL
Authentication authority asserted identity
Medium Mandatory Level

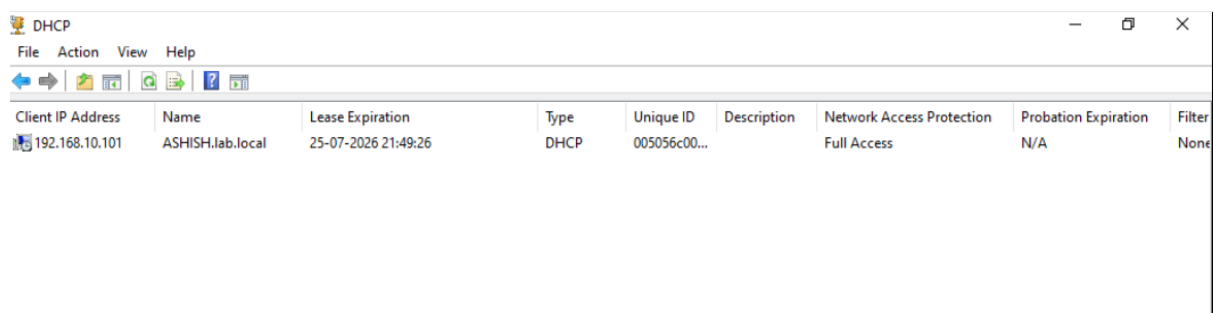
C:\Users\ashish>
```

**Figure 16.1** *gpresult /r* showing applied Group Policies

```
C:\Users\ashish>nslookup dc01.lab.local
Server:  DC01.lab.local
Address:  192.168.10.10

Name:     dc01.lab.local
Address:  192.168.10.10
```

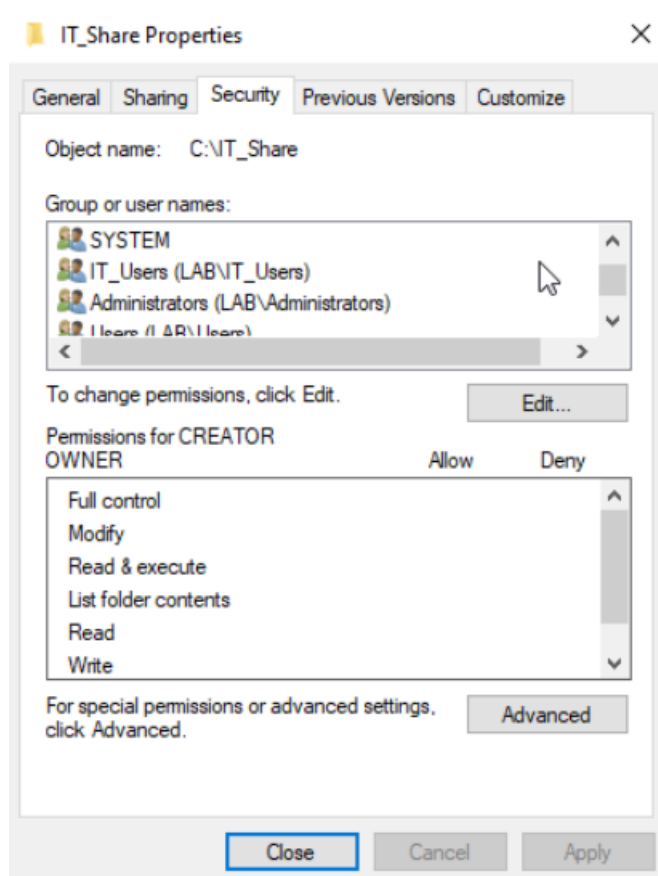
*Figure 16.2 nslookup dc01.lab.local after fixing DNS*



The screenshot shows the DHCP console window with a single lease entry for CLIENT01. The table below represents the data shown in the console.

| Client IP Address | Name             | Lease Expiration    | Type | Unique ID    | Description | Network Access Protection | Probation Expiration | Filter |
|-------------------|------------------|---------------------|------|--------------|-------------|---------------------------|----------------------|--------|
| 192.168.10.101    | ASHISH.lab.local | 25-07-2026 21:49:26 | DHCP | 005056c00... |             | Full Access               | N/A                  | None   |

*Figure 16.3 DHCP Address Leases showing CLIENT01 received an IP address*



*Figure 16.4 Shared Folder permissions (Security or Sharing tab) after correcting NTFS permissions*

## **Future Enhancements**

While the current implementation successfully provides a functional enterprise network environment, several additional features and technologies can be integrated in the future to improve scalability, security, automation, and fault tolerance.

## **Conclusion**

This project successfully demonstrated the design, implementation, and validation of a Windows Server 2022–based enterprise network integrated with Cisco networking technologies. A complete lab environment was built using VMware Workstation Pro and Cisco Packet Tracer to simulate a real-world organizational infrastructure.

The implementation included the deployment of Active Directory Domain Services (AD DS), Domain Name System (DNS), Dynamic Host Configuration Protocol (DHCP), Group Policy Objects (GPOs), shared folder management with NTFS permissions, and centralized user administration. On the networking side, VLAN segmentation, IEEE 802.1Q trunking, and Router-on-a-Stick inter-VLAN routing were configured to provide logical network separation and communication between different network segments.

Comprehensive testing and validation confirmed that all core services operated correctly. Domain users were able to authenticate successfully, clients received IP addresses automatically through DHCP, DNS resolved hostnames correctly, Group Policies were enforced, and authorized users could securely access shared resources. Throughout the project, configuration issues were systematically identified and resolved using standard troubleshooting techniques and Windows administrative tools.

## **References**

1. Microsoft Learn. *Windows Server Documentation*. Available at: <https://learn.microsoft.com/windows-server/>
2. Microsoft Learn. *Active Directory Domain Services (AD DS)*. Available at: <https://learn.microsoft.com/windows-server/identity/ad-ds/>
3. Microsoft Learn. *DNS Overview*. Available at: <https://learn.microsoft.com/windows-server/networking/dns/>
4. Microsoft Learn. *DHCP Overview*. Available at: <https://learn.microsoft.com/windows-server/networking/technologies/dhcp/>
5. Microsoft Learn. *Group Policy Documentation*. Available at: <https://learn.microsoft.com/windows-server/identity/ad-ds/manage/group-policy/>
6. Microsoft Learn. *File Sharing and Storage*. Available at: <https://learn.microsoft.com/windows-server/storage/>
7. Cisco Networking Academy. *Cisco Packet Tracer*. Available at: <https://www.netacad.com/>
8. Cisco. *Cisco IOS Configuration Guides*. Available at: <https://www.cisco.com/>
9. VMware. *VMware Workstation Pro Documentation*. Available at: <https://docs.vmware.com/>
10. RFC 1918. *Address Allocation for Private Internets*. Internet Engineering Task Force (IETF). Available at: <https://www.rfc-editor.org/rfc/rfc1918>
11. RFC 2131. *Dynamic Host Configuration Protocol (DHCP)*. Internet Engineering Task Force (IETF). Available at: <https://www.rfc-editor.org/rfc/rfc2131>
12. RFC 1034 and RFC 1035. *Domain Names – Concepts, Facilities, and Implementation*. Internet Engineering Task Force (IETF). Available at: <https://www.rfc-editor.org/>

This project provided practical experience in Windows Server administration, Active Directory management, enterprise networking, network security, and troubleshooting. It strengthened skills in deploying and managing enterprise infrastructure while demonstrating the ability to integrate Microsoft server technologies with Cisco networking concepts. The completed environment serves as a strong demonstration of practical system administration and network engineering capabilities applicable to real-world enterprise environments.